Fundamentos de Aprendizagem Automática

Class 07 — Ensemble Models

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One Model at a Time

Where We Left Off

linear, probabilistic
Classes 03–04

margin, network
Class 05

tree, neighbours
Class 06

Class 07: many models, *one* decision

- Until now: **one** model, tuned
- Today: first see where one model fails, then combine models

Roadmap of the Class

Part	Question	Methods
Shallow models	where does one model fail?	six models, 17 sets
Why combining helps	when does a crowd win?	majority, mean
Voting	how to combine by hand?	hard, soft, weighted, k-NN
Bagging	how to cut variance?	bagged trees, random forest
Boosting	how to cut bias?	AdaBoost, gradient boosting
Meta models	how to learn the mix?	stacking, neural stacking
Tuning	how to set the knobs?	random search, Optuna
Real tables	which family wins?	11 tables, ranks
Is the gain real?	is it luck?	McNemar, permutation, Wilcoxon

Notation N7: Members

Each symbol is new in this class; earlier symbols keep their meaning.

symbol	meaning	shape	example
B	number of members	scalar	$B = 100$ trees
b	index of a member or round	scalar	$b = 1, \dots, B$
$\hat{f}_b$	member b	function	one tree
$\hat{f}_{\text{ens}}$	the ensemble	function	the vote
$\hat{p}_b(k \mid x)$	probability member b gives to class k	scalar	0.8
u_b	weight of member b in a vote	scalar	0.316
η	shrinkage (the learning rate of Class 02)	scalar	0.1

Notation N7b: Voters, Variance, Weights

symbol	meaning	shape	example
γ	chance that one voter is right	scalar	0.6
a_B	chance that the majority of B voters is right	scalar	$a_3 = 0.648$
v, V_B	variance of one member, of the mean of B	scalar	$V_5 = 0.680$
ρ	correlation of two members' errors	scalar	$\rho = 0.6$
ξ	share of examples left out of a bootstrap sample	scalar	0.368
$w_i^{(b)}$	weight of example i in round b	scalar	$1/n$ at the start
err_b	weighted error of member b	scalar	0.20
β_b	weight of member b in AdaBoost	scalar	0.693
E_b, Z_b	exponential loss of member b, weight normaliser	scalar	0.800

Notation N7c: Examples and Meta Models

symbol	meaning	shape	example
n	number of examples (Class 01)	scalar	600
x_i	example i (Class 01)	vector	one point of a plane
y_i	targets (Class 01)	scalar	$+1$ or -1
$\hat{y}$	prediction	scalar	0.8
z_A	last hidden layer of network A	vector	128 values
z_B	last hidden layer of network B	vector	64 values

Notation N7d: Scores

symbol	meaning	shape	example
TP	confusion-matrix counts (Class 01)	scalar	30
TN	confusion-matrix counts (Class 01)	scalar	50
FP	confusion-matrix counts (Class 01)	scalar	10
FN	confusion-matrix counts (Class 01)	scalar	10
G	the product under the root of MCC	scalar	5760000
MCC	Matthews correlation coefficient	scalar	0.583

Shallow Models

A **shallow model** is one fitted model that combines nothing.

logistic, or linear
Class 03

naive Bayes, or polynomial
Classes 03–04

tree, depth 5
Class 06

1-NN
Class 06

SVM, RBF kernel
Class 05

MLP, one hidden layer
Class 05

- Nearest neighbours (NN) with $k > 1$ are left out: k-NN is already a vote

Two Scores: MCC and MAE

Matthews correlation (MCC) for classes, **mean absolute error** (MAE) for values.

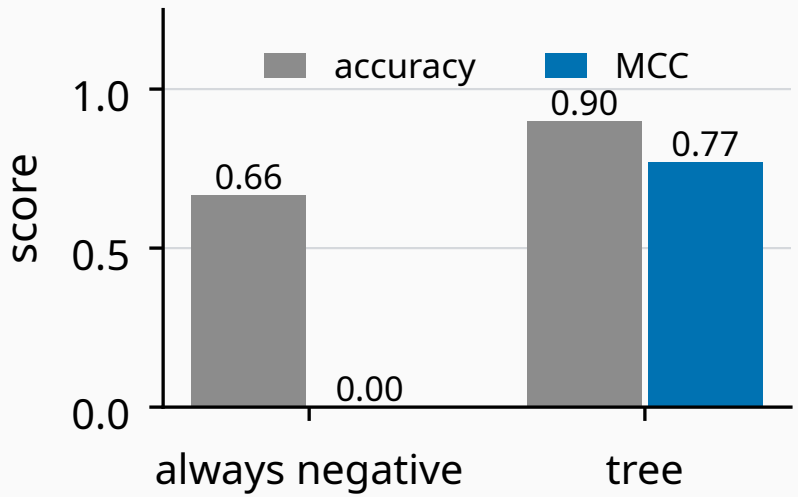


$$G = (TP+FP)(TP+FN)(TN+FP)(TN+FN) \quad \text{(definition)}$$

$$\text{MCC} = \frac{TP\,TN - FP\,FN}{\sqrt{G}} \quad \text{(definition)}$$

- Here: $\sqrt{G} = 2400$, so MCC $= 1400/2400 = 0.583$
- MAE: the mean of $|y_i - \hat{y}_i|$, in the units of y

Why Not Accuracy?



- A rare class: 0.335 of the points
- Always negative: accuracy 0.665, MCC 0
- MCC 1 is perfect, 0 is a coin
- Errors 0.1, 0.1, 0.1, 0.1, 2: MAE 0.48, root mean squared error (RMSE) 0.90

The Data Sets

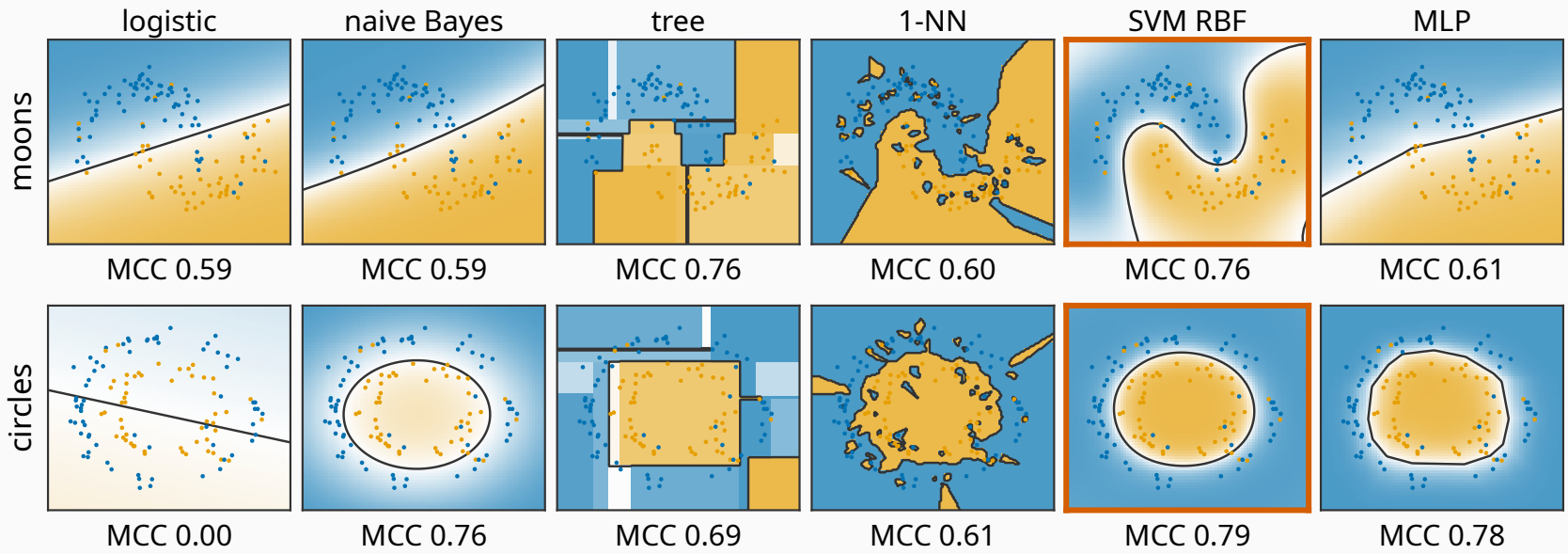
Generated sets with default models show **where a model breaks**, not which is best.

group	sets	train
planes	moons, circles, XOR, skewed blobs, spiral, checkerboard	600
curves	cubic, sine, exponential, chirp, step, sawtooth	120
tables	20 features, Hastie, kinked border, two Friedman sets	1000

- 10 % flipped labels on four planes: MCC stops at **0.80**
- Noise of sd 0.10 on the curves: MAE stops at **0.08**

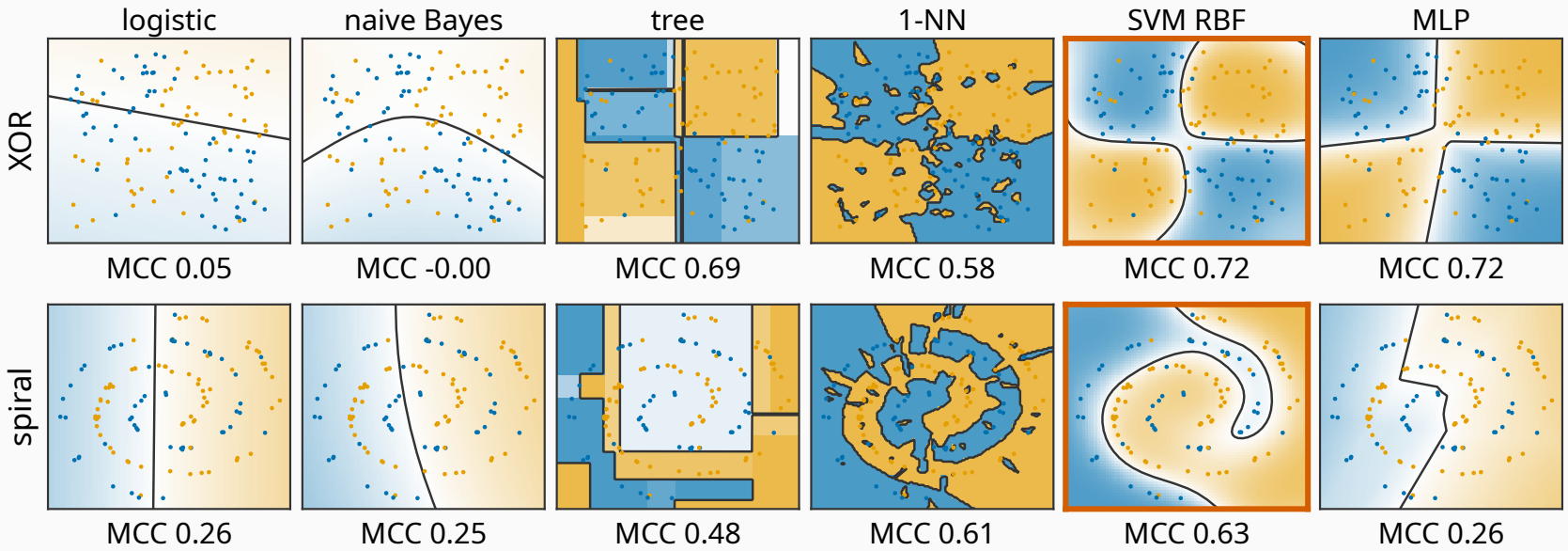
Curved Data Breaks Straight Frontiers

Colour: probability of class 1. Red frame: best model of the row.



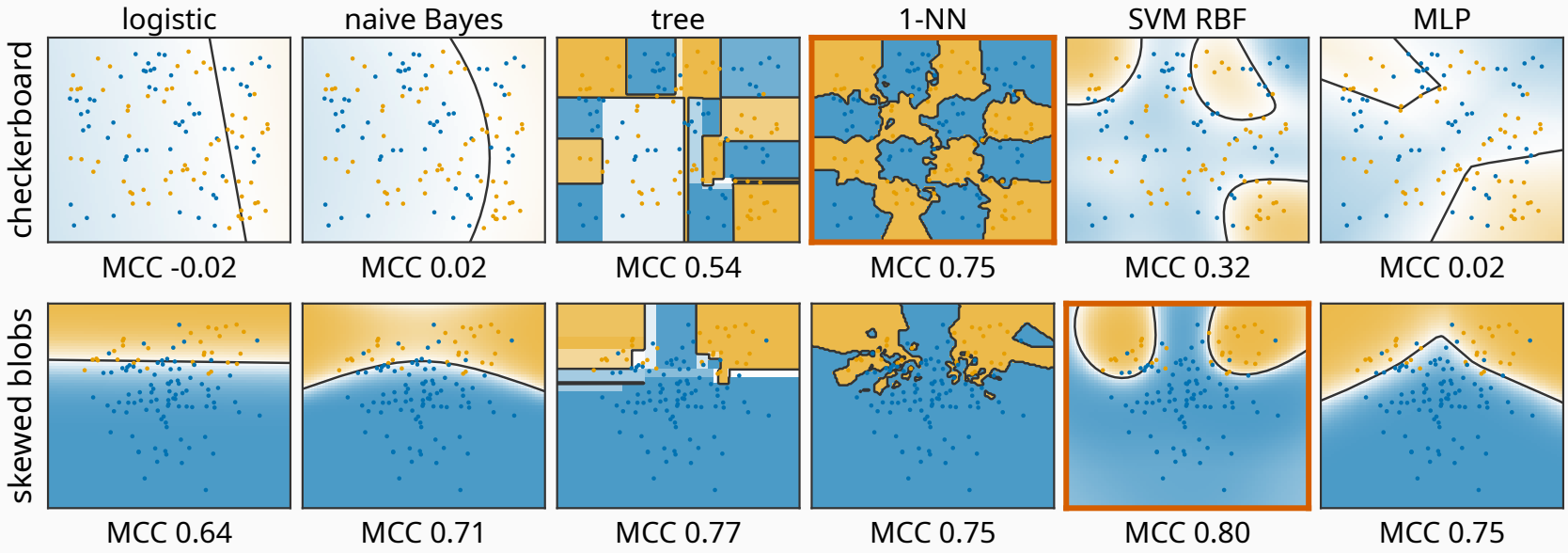
- Logistic regression: MCC 0.00 on circles, **a line cannot close a loop**
- 1-NN draws islands around the flipped labels

Two Regions per Class Break the Rest



- Exclusive or (XOR): logistic and naive Bayes are at MCC 0.05 and 0.00
- Spiral: best MCC 0.63, far below the ceiling of 0.80
- Support vector machine (SVM), multilayer perceptron (MLP): too smooth here

Small Cells and a Rare Class



- Checkerboard: only 1-NN follows the cells (0.75)
- Skewed blobs: the models are close, 0.64 to 0.80

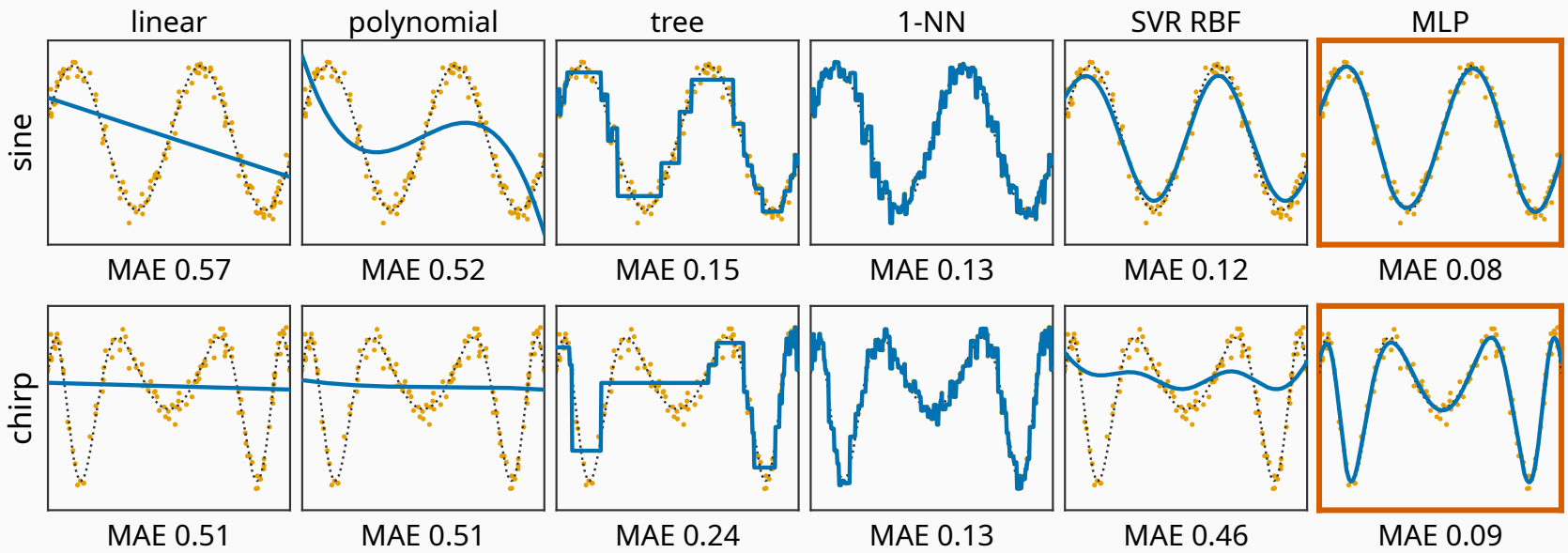
The Board of Classification

Mean test MCC of five seeds. Grey cells fill up in the next parts.

	logistic	naive Bayes	tree	1-NN	SVM RBF	MLP	hard vote	soft vote	weighted vote	15-NN	15-NN by distance	bagged trees	random forest	AdaBoost	gradient boosting	XGBoost	stacking	NN A	NN B	neural stack
moons	.60	.59	.73	.62	**.76**	.70														
circles	.04	.74	.71	.63	**.79**	.78														
XOR	.00	-.04	.58	.58	.71	**.71**														
skewed blobs	.64	.72	.77	.75	**.79**	.76														
spiral	.24	.23	.43	**.63**	.52	.31														
checkerboard	.05	-.00	.37	**.75**	.25	.11														
20 features	.38	.36	.40	.52	**.67**	.61														
Hastie	.02	.89	.29	.38	**.90**	.72														
kinked border	.60	.67	.55	.56	.74	**.78**														

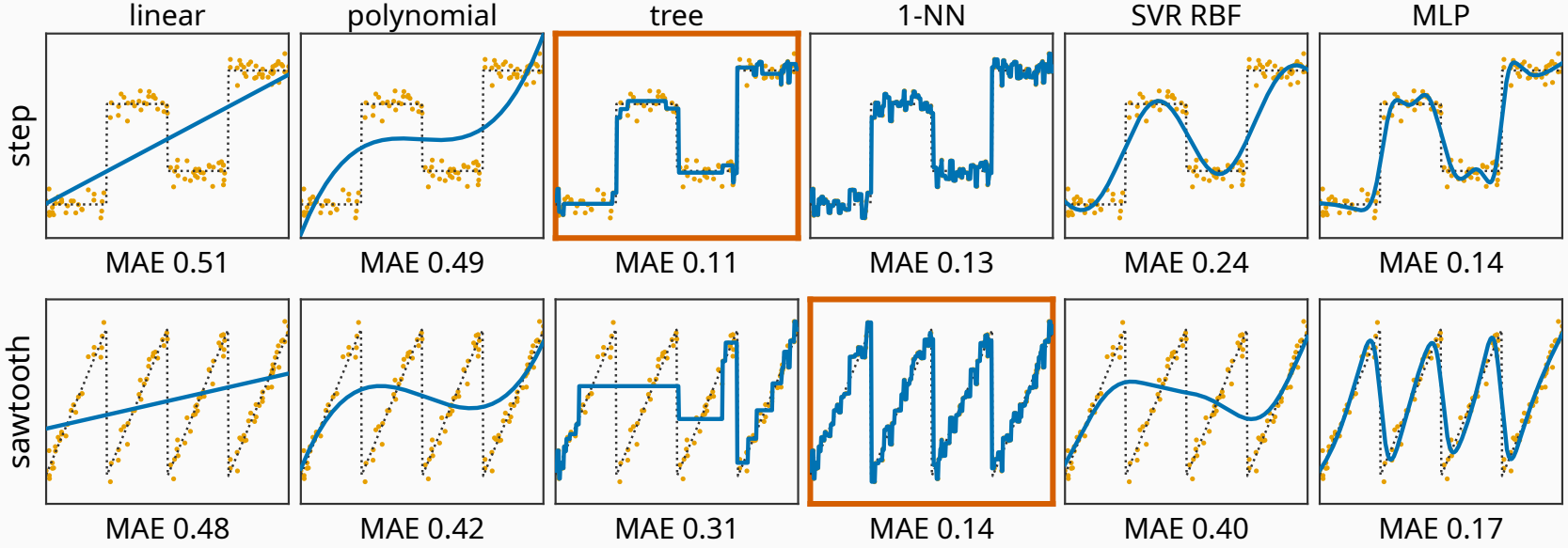
- **Each model has a set that breaks it**; none wins everywhere

Oscillations Break Global Models



- Degree 3 has too few bends: MAE 0.52 on the sine
- Support vector regression (SVR) misses the fast chirp: MAE 0.46

Jumps Break Smooth Models



- Step: the tree wins with 0.11, smooth models round the corners
- Sawtooth: **no model is near 0.08**, the best is 1-NN with 0.14

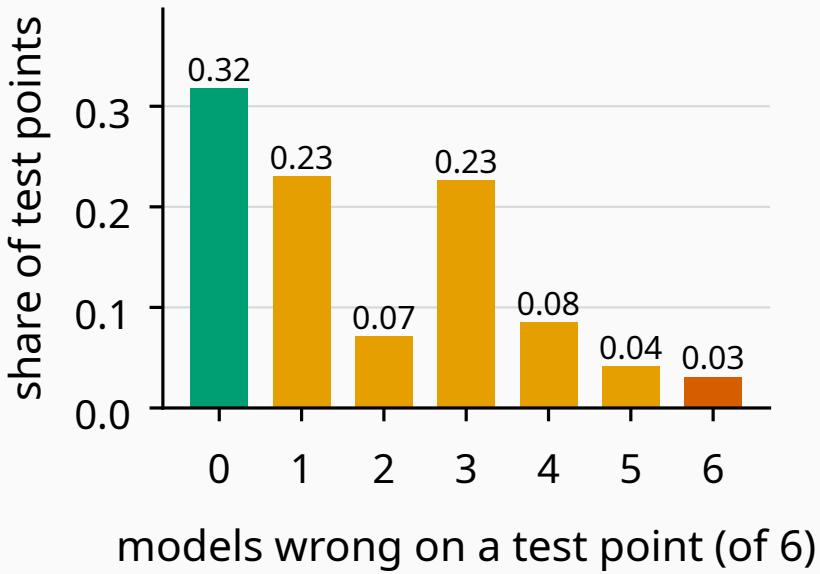
The Board of Regression

Mean test MAE of five seeds. The colour compares a cell with the best of its row.

	linear	polynomial	tree	1-NN	SVR RBF	MLP	median vote	mean vote	weighted vote	5-NN	5-NN by distance	bagged trees	random forest	AdaBoost	gradient boosting	XGBoost	stacking	NN A	NN B	neural stack
cubic	.88	**.08**	.17	.13	.17	.09														
sine	.57	.50	.14	.12	.11	**.08**														
exponential	.42	.09	.10	.12	.10	**.08**														
chirp	.51	.52	.18	.13	.41	**.10**														
step	.52	.49	**.11**	.13	.23	.15														
sawtooth	.49	.43	.28	**.14**	.40	.17														
Friedman 1	1.9	**.40**	2.1	2.7	1.3	.50														
Friedman kinks	2.6	.97	2.9	3.0	1.9	**.73**														

- Friedman 1: polynomial 0.40, the MLP 0.50, 1-NN 2.72
- The ranking changes from row to row, as in classification

Do the Errors Overlap?



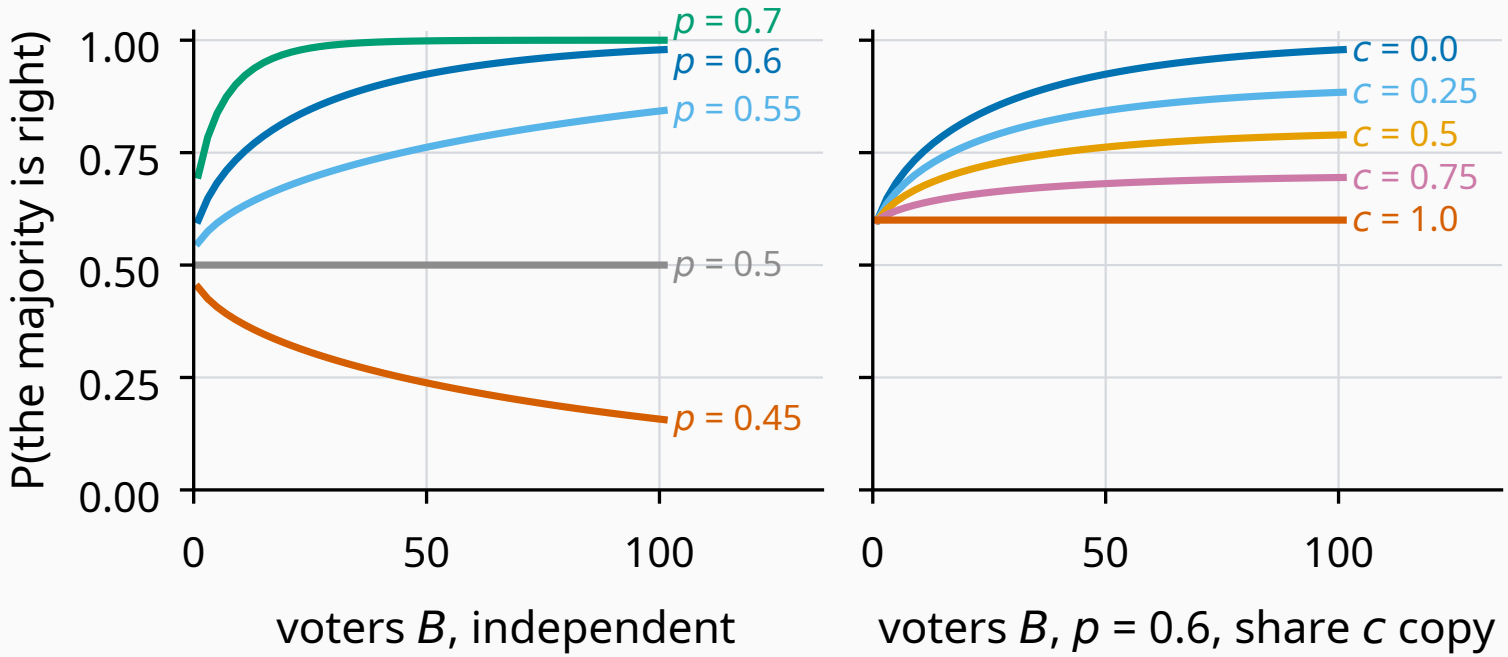
- 0.317 of the points: right for all six
- 0.030: wrong for all six
- **The models disagree on 0.652**
- That is the room a combination can use

DEMO E1: `01_shallow_models.ipynb`

Why Combining Helps

The Crowd Wins If Voters Are Independent

B voters, each right with chance γ; the majority is right with the binomial sum.



- $\gamma = 0.6$: $B = 3$ gives 0.648, $B = 11$ gives 0.753, $B = 101$ gives 0.979

Condorcet for Three Voters

$$
\begin{aligned}
a_3 &= \gamma^3 + 3\gamma^2(1-\gamma) && \text{(definition)} \\
&= 0.6^3 + 3 \cdot 0.6^2 \cdot 0.4 && \text{(substitute } \gamma = 0.6\text{)}
\end{aligned}
$$

3 right: 0.216

2 right: 0.432

1 or 0 right: 0.352

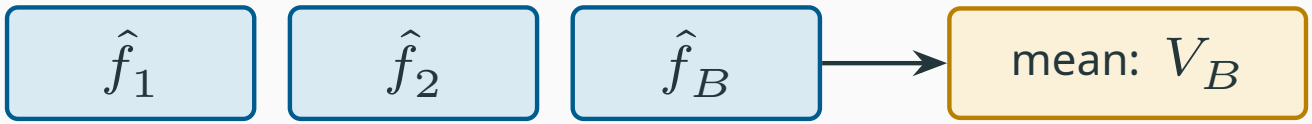
- Assumption: the voters err **independently**
- Copies do not help: if half the voters copy, $B = 101$ gives 0.790

Averaging Reduces Variance

For regression: the mean of B predictions, each with variance v and pairwise correlation ρ:

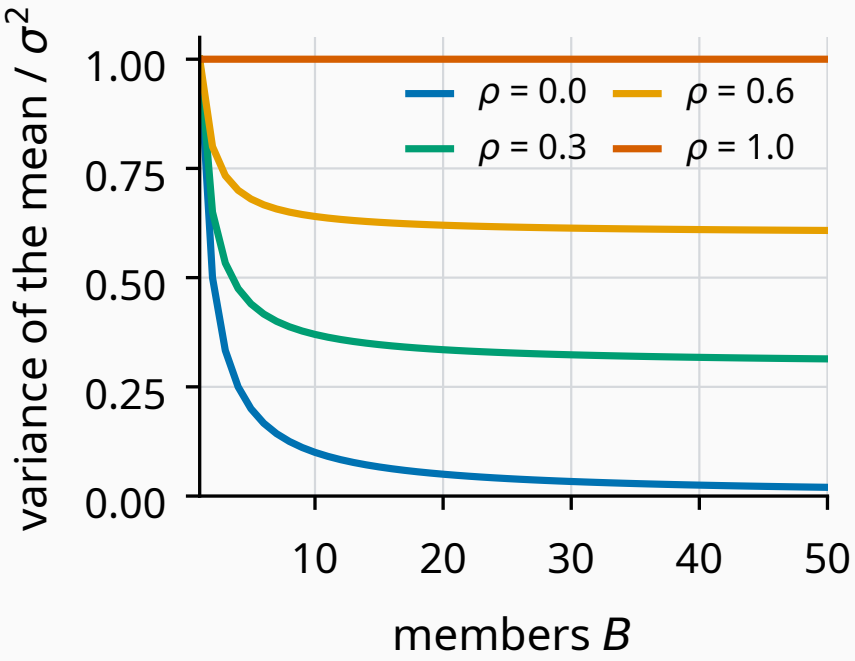
$$V_B = \frac{B\,v + B(B-1)\,\rho\,v}{B^2} \quad \text{(definition)}$$

$$= \rho\,v + \frac{(1-\rho)\,v}{B} \quad \text{(simplify)}$$



- With $v = 1$, $\rho = 0.6$: $B = 5$ gives 0.680, $B = 50$ gives 0.608

The Floor Is the Correlation



- $\rho = 0$, $B = 50$: variance 0.020; $\rho = 0.3$: 0.314; $\rho = 0.6$: 0.608
- More members cannot remove the shared part ρv
- **Diverse members matter more than many members**

Four Families of Ensembles

Voting
fit different models,
combine the outputs

Bagging
one learner on resamples,
in parallel: cut *variance*

Boosting
members in sequence,
each fixes the last: cut *bias*

Meta models
a model learns the combination

- Same data, same scores: each family adds a row to the same figures

Voting

Hard Vote, Soft Vote, Weighted Vote

The members are the **six shallow models**, each fitted on the whole training set.

Hard: each member names a class, the majority wins

Soft: average the probabilities, take the largest

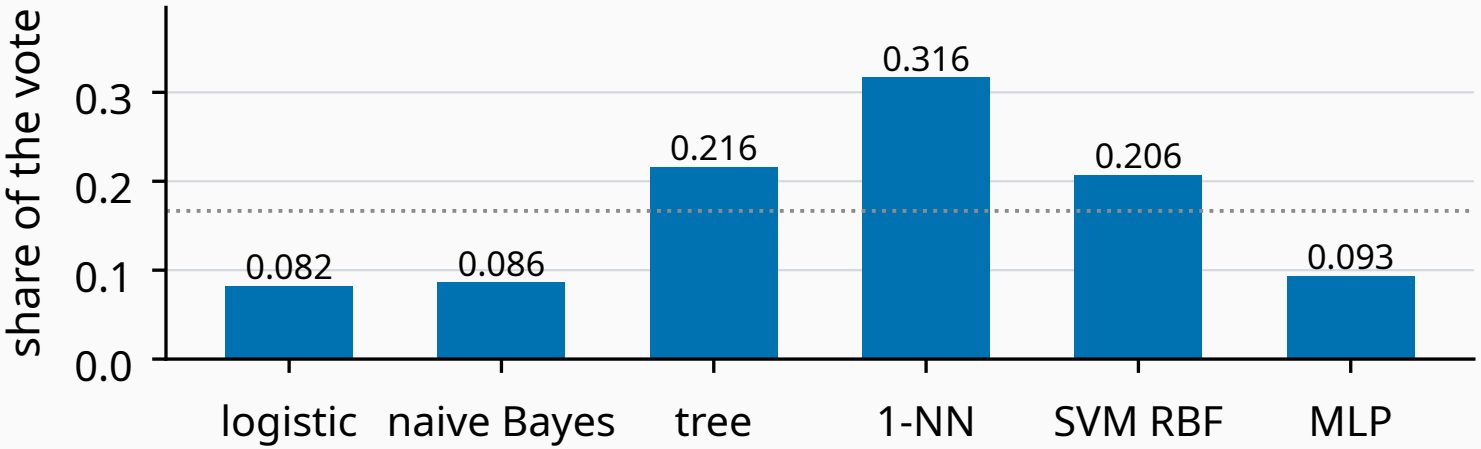
Weighted: a soft vote, better members count more

- Regression: the **median** plays the hard vote, the **mean** the soft vote

$$\hat{f}_{\text{ens}}(x) = \arg\max_{k} \sum_{b} u_b \, \hat{p}_b(k \mid x) \quad \text{(definition)}$$

Where the Weights Come From

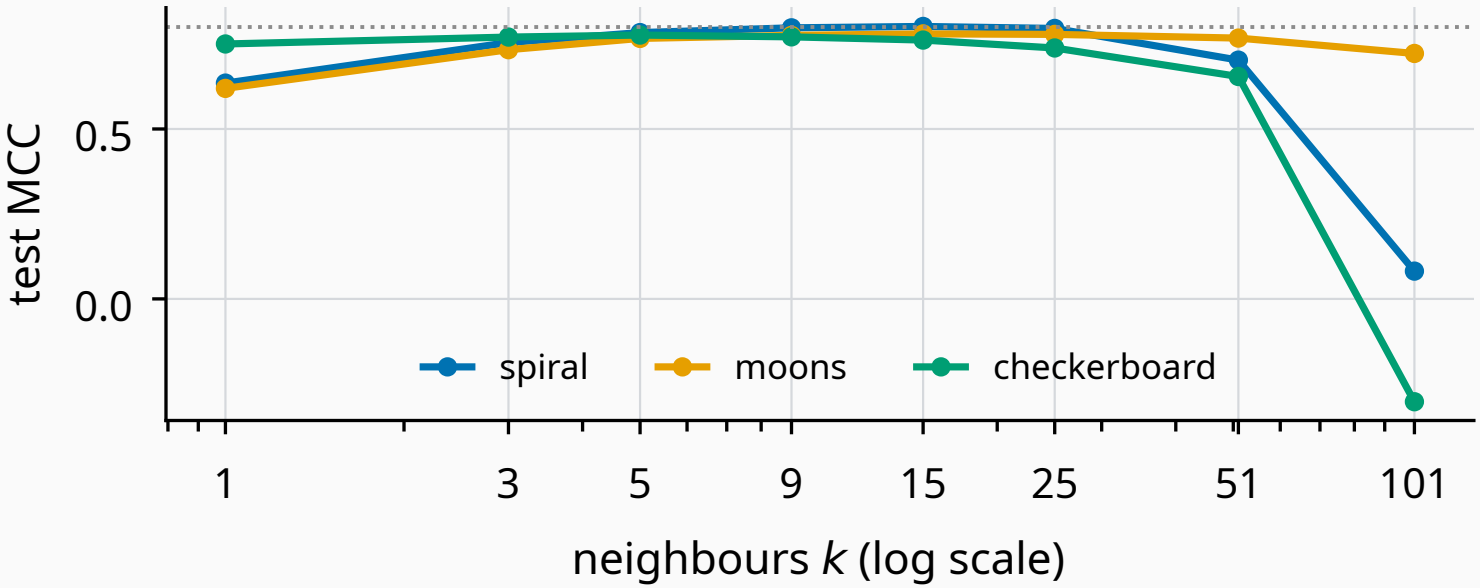
u_b: the MCC of member b in 5-fold cross-validation on the training set, floored at 0.



- Spiral: 1-NN gets 0.316, logistic 0.082; dotted: equal votes
- Regression: u_b is the inverse of the cross-validated MAE

k-NN Is a Vote

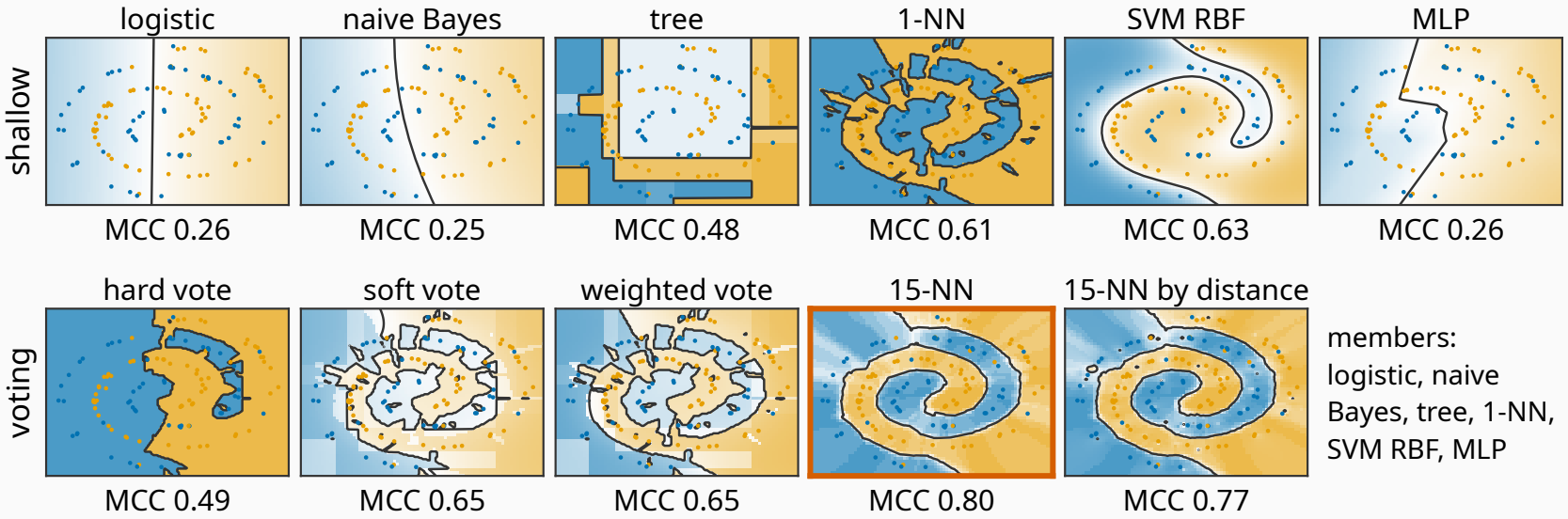
k-NN with $k > 1$ is a vote of the k nearest points: each neighbour is one voter.



- Spiral: $k = 1$ gives 0.635, $k = 15$ gives **0.802**, $k = 101$ gives 0.082
- **Weighted k-NN:** a closer neighbour counts more
- Too many neighbours vote across the frontier

Voting on the Spiral

The vote panels say which shallow models they use.



- 15-NN follows the spiral and ignores the flipped labels (0.80)
- The soft vote (0.65) beats each member

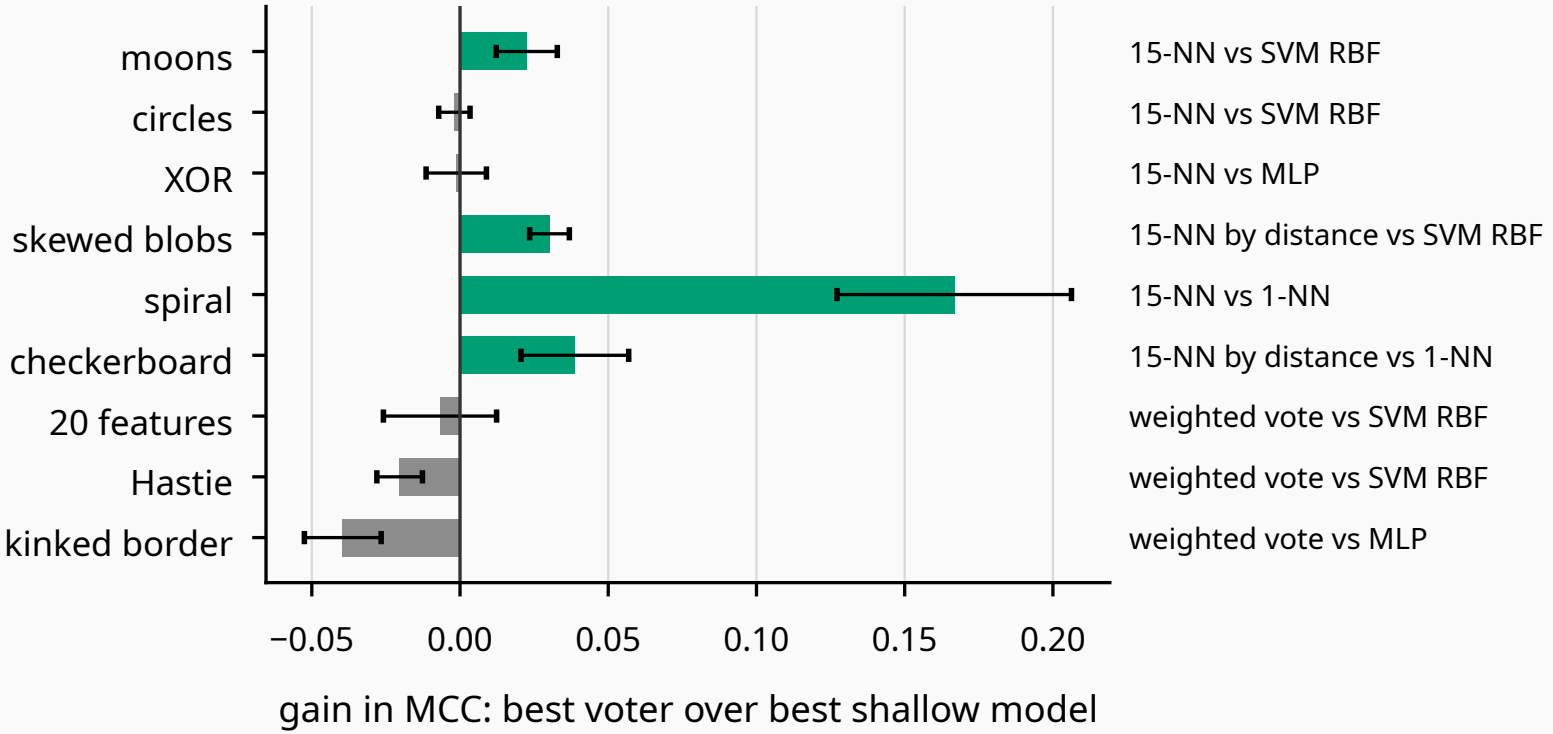
Voting Fills Five Columns

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moons	.60	.59	.73	.62	.76	.70	.72	.74	.74	**.78**	.77									
circles	.04	.74	.71	.63	**.79**	.78	.78	.77	.77	.79	.78									
XOR	.00	-.04	.58	.58	.71	**.71**	.58	.69	.70	.71	.71									
skewed blobs	.64	.72	.77	.75	.79	.76	.78	.79	.79	.79	**.82**									
spiral	.24	.23	.43	.63	.52	.31	.51	.64	.66	**.80**	.79									
checkerboard	.05	-.00	.37	.75	.25	.11	.30	.74	.75	.76	**.79**									
20 features	.38	.36	.40	.52	**.67**	.61	.60	.64	.66	.61	.61									
Hastie	.02	.89	.29	.38	**.90**	.72	.69	.81	.88	.25	.25									
kinked border	.60	.67	.55	.56	.74	**.78**	.72	.74	.75	.66	.67									

- Spiral 0.802, skewed blobs 0.817, moons 0.780: new best scores
- Tables: the votes stay below the SVM

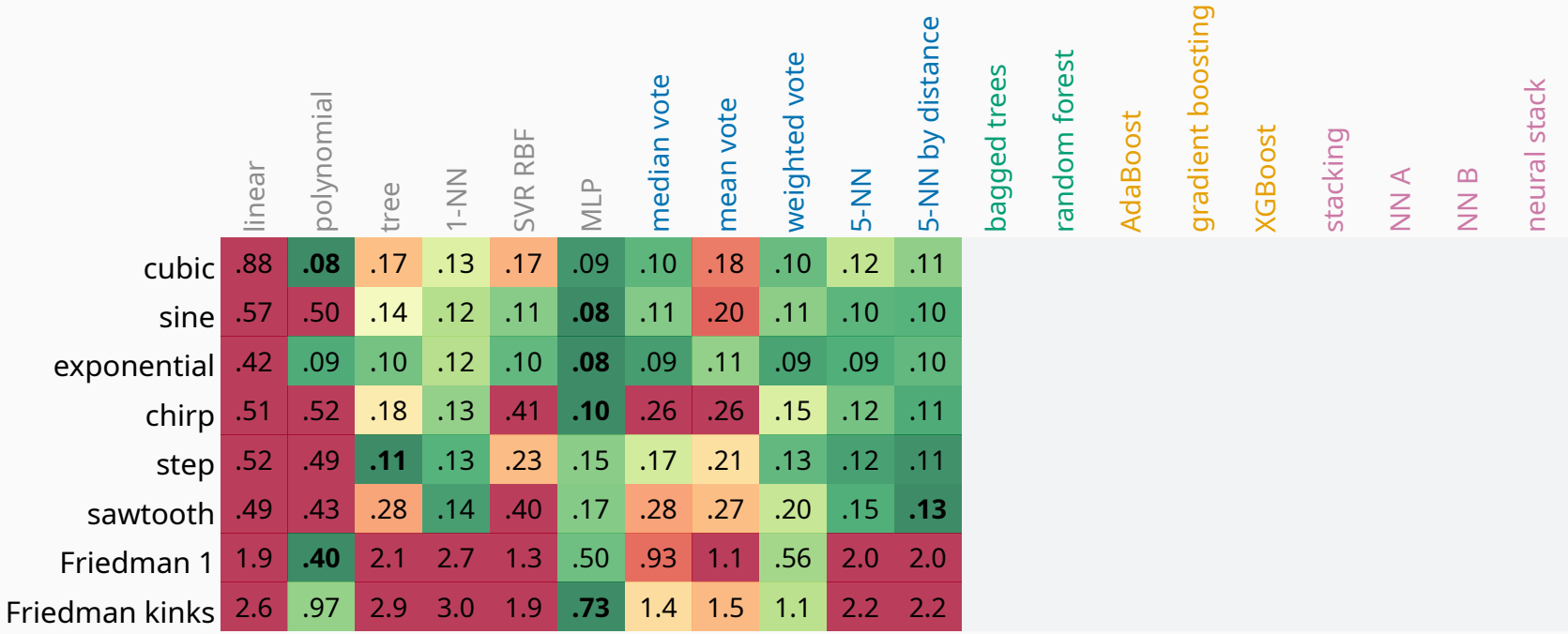
Is the Voting Gain Real?

Gain: best voter minus best shallow model. Green: won on all five seeds.



- **Four wins of nine:** spiral $+0.167$, checkerboard $+0.039$, skewed blobs $+0.030$, moons $+0.023$
- Circles and XOR sit at the ceiling: nothing is left to gain

Voting in Regression



	linear	polynomial	tree	1-NN	SVR RBF	MLP	median vote	mean vote	weighted vote	5-NN	5-NN by distance	bagged trees	random forest	AdaBoost	gradient boosting	XGBoost	stacking	NN A	NN B	neural stack
cubic	.88	**.08**	.17	.13	.17	.09	.10	.18	.10	.12	.11									
sine	.57	.50	.14	.12	.11	**.08**	.11	.20	.11	.10	.10									
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Friedman 1	1.9	**.40**	2.1	2.7	1.3	.50	.93	1.1	.56	2.0	2.0									
Friedman kinks	2.6	.97	2.9	3.0	1.9	**.73**	1.4	1.5	1.1	2.2	2.2									

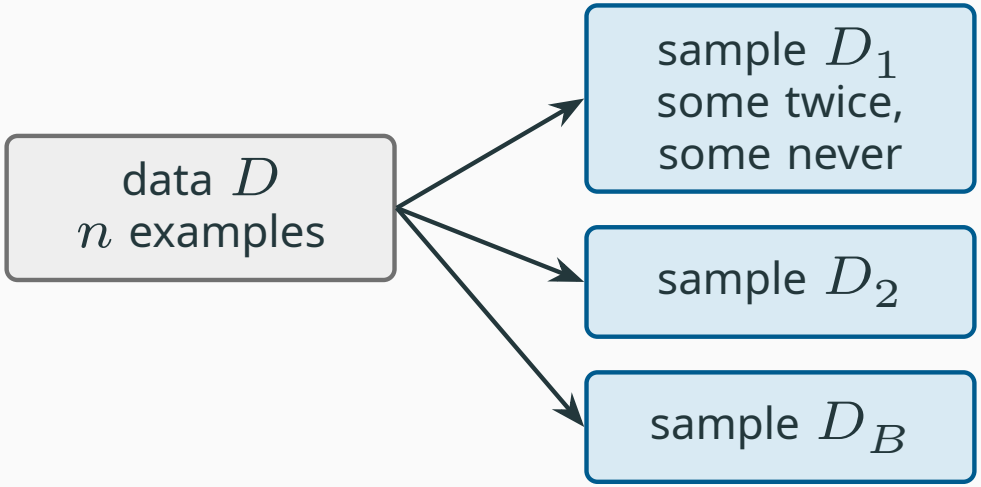
- The MLP is at the floor on smooth curves: the vote can only lose
- Only the sawtooth gains: 5-NN by distance 0.132 against 0.142
- **A vote cannot beat a member that is already right**

DEMO E2: `02_voting.ipynb`

Bagging

The Bootstrap Sample

To get B different training sets from one, draw n examples **with replacement**.

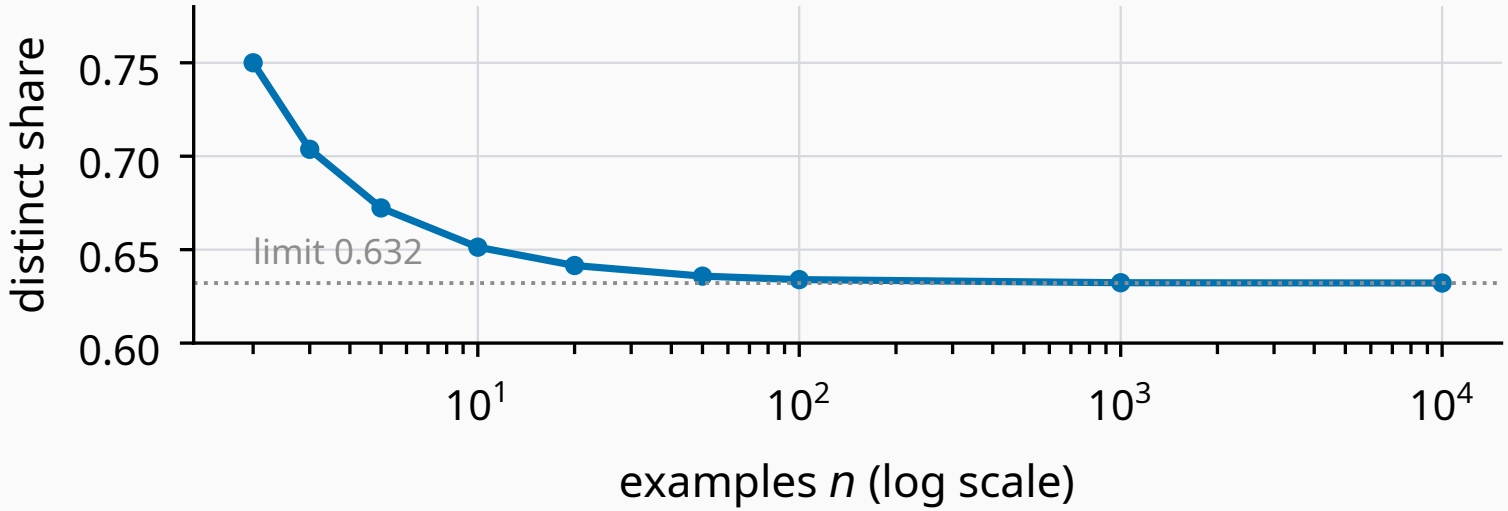


- Each sample holds about two thirds of the distinct examples

How Many Distinct Examples?

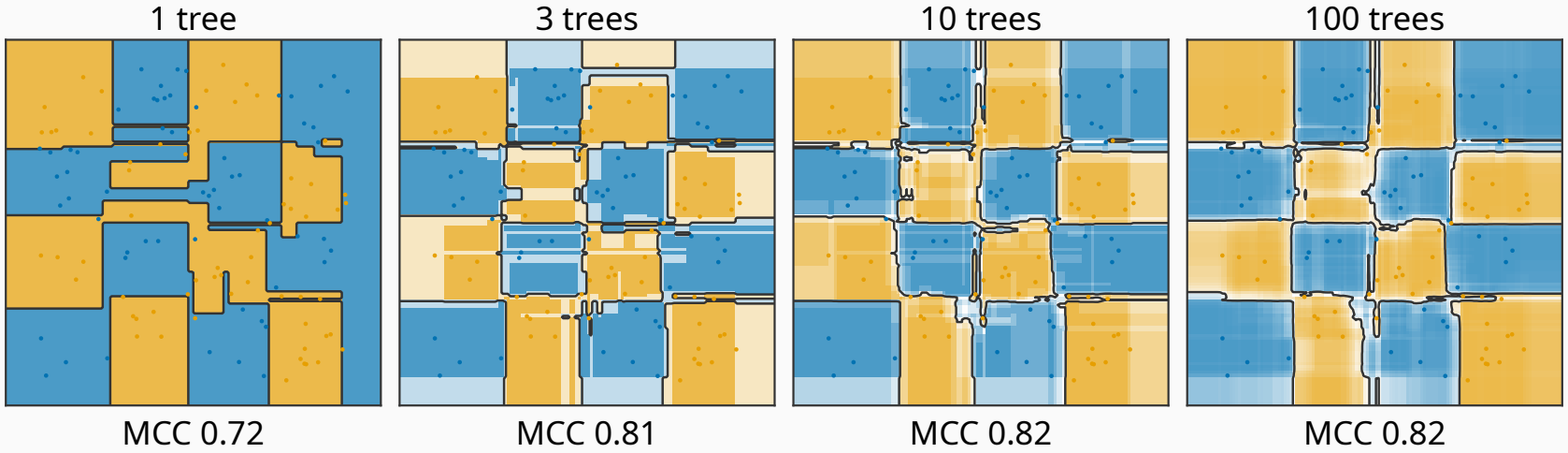
$$\xi = \left(1 - \frac{1}{n}\right)^n \quad \text{(definition)}$$

$$\xi \to e^{-1} \quad \text{(identity: limit)}$$



- $\xi = 0.368$: all n draws miss example i
- That left-out share is the **out-of-bag** (OOB) set of the member

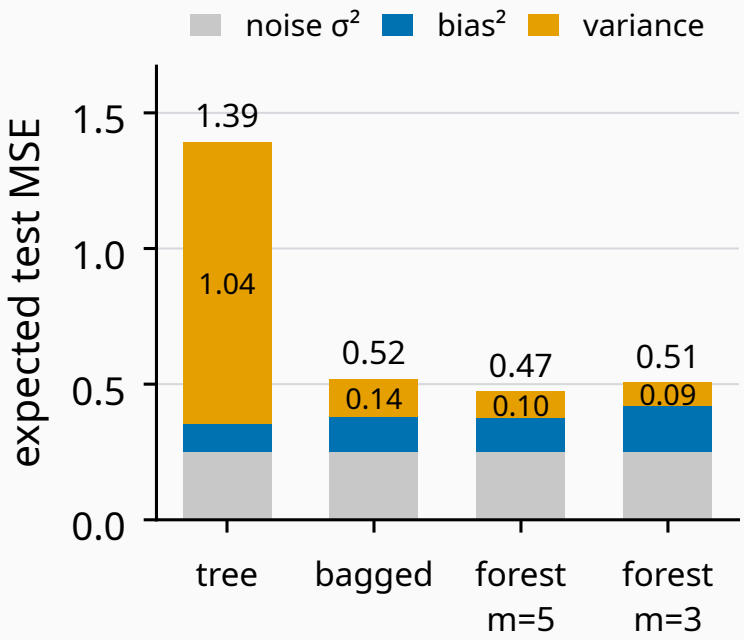
Bagging in Action: the Checkerboard



- One deep tree 0.72; $B = 3, 10, 100$: MCC 0.81, 0.82, 0.82
- `BaggingClassifier` with 100 trees: 0.830; by hand: 0.819

Variance Cut, Bias Kept

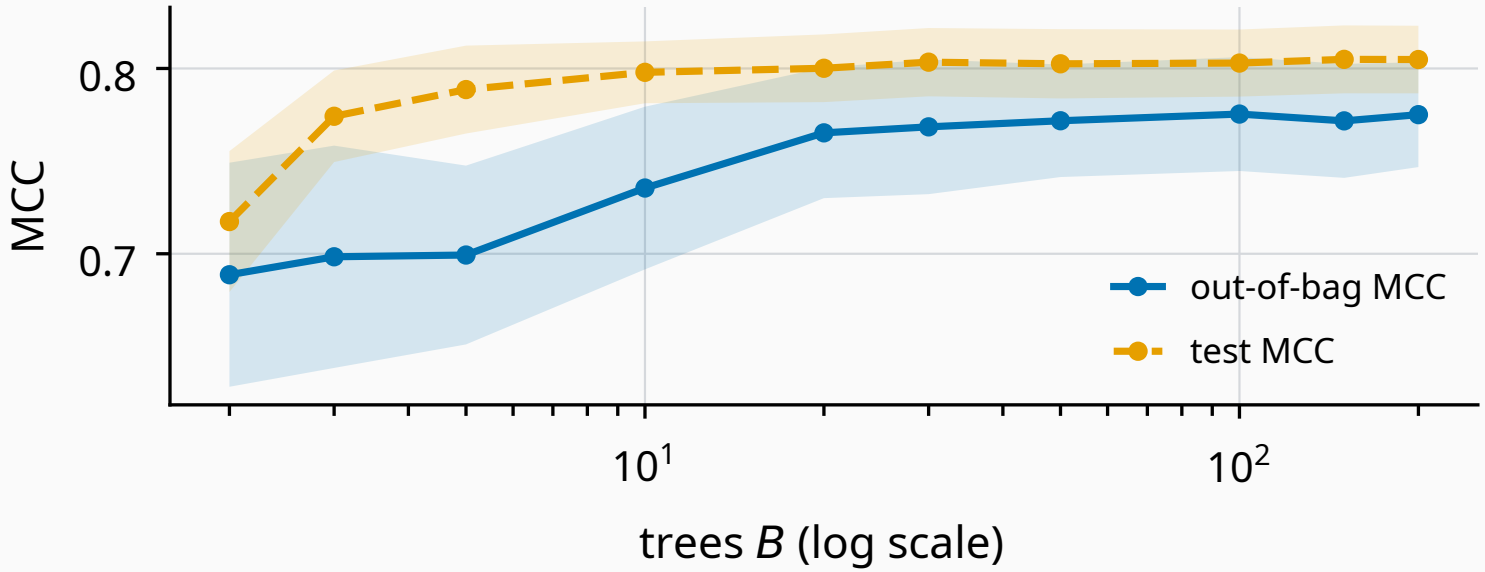
The decomposition needs the squared error: 200 training sets, noise 0.25.



- One tree: bias^2 0.105, variance 1.038
- Bagged: 0.131, 0.140; forest ($m = 5$): 0.125, 0.097
- Variance falls by a factor 7; the bias barely moves

Out-of-Bag Score: Free Validation

Predict each example with the trees that did **not** train on it.

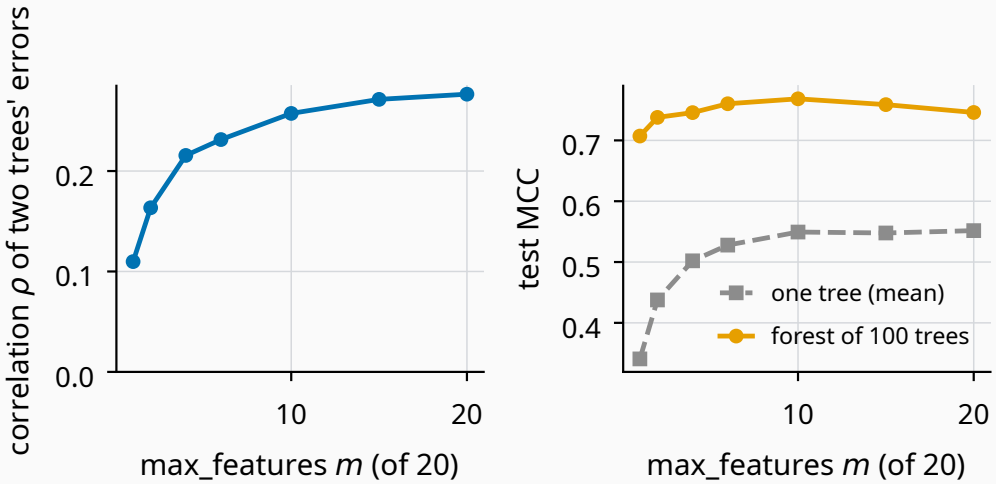


- $B = 5, 10, 100$: OOB 0.699, 0.735, 0.775; test 0.789, 0.798, 0.803
- OOB is **pessimistic**: each vote uses a third of the trees

From Bagging to a Random Forest

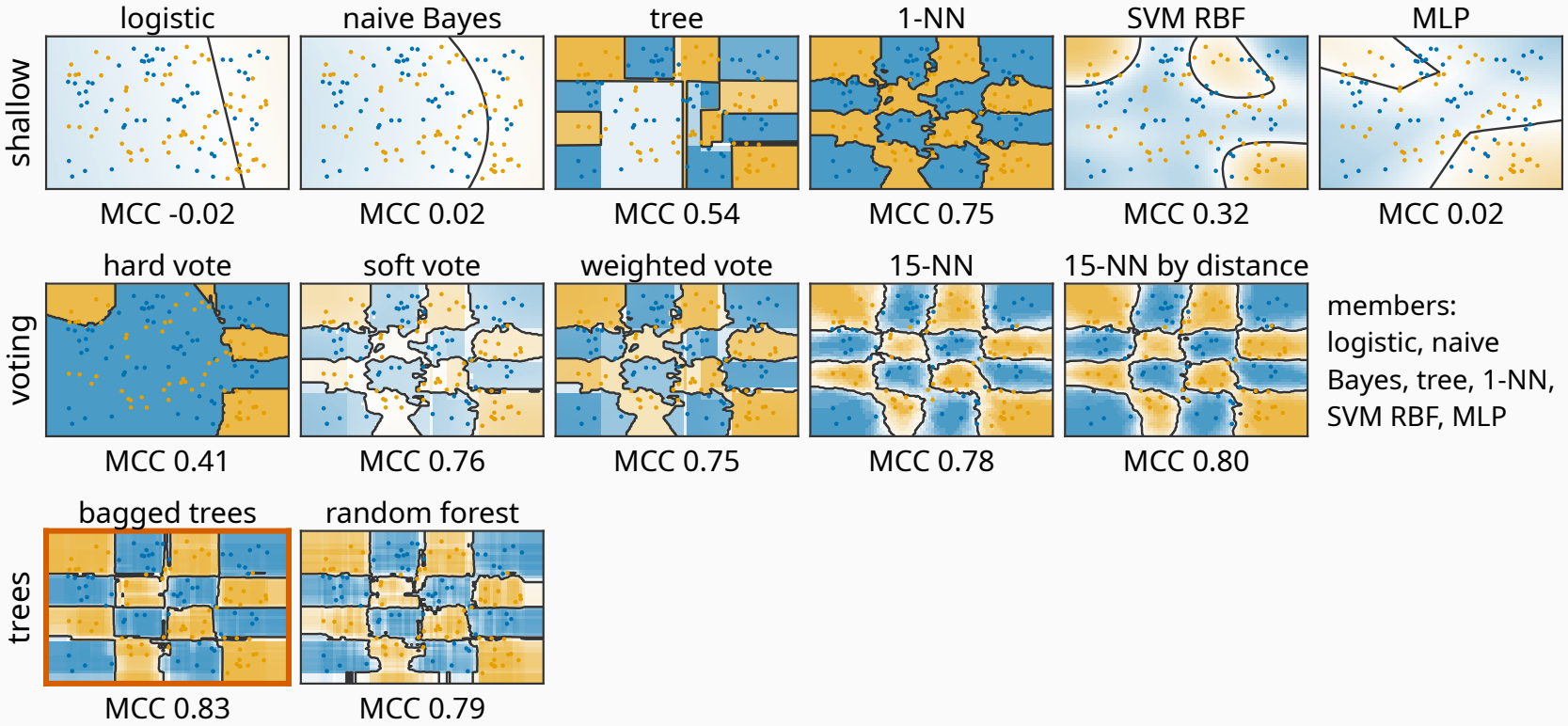
Bagged trees are correlated: they all pick the same strong feature first.

Random forest: at each split consider only m of the d features.



- $m = 1$: $\rho = 0.110$, 0.707; $m = 10$: 0.768; $m = 20$: $\rho = 0.277$, 0.746

Bagging Joins the Gallery



- Bagged trees reach **0.83**, the best so far: squares suit straight cuts

Bagging Fills Two Columns

	logistic	naive Bayes	tree	1-NN	SVM RBF	MLP	hard vote	soft vote	weighted vote	15-NN	15-NN by distance	bagged trees	random forest	AdaBoost	gradient boosting	XGBoost	stacking	NN A	NN B	neural stack
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circles	.04	.74	.71	.63	**.79**	.78	.78	.77	.77	.79	.78	.71	.72							
XOR	.00	-.04	.58	.58	.71	**.71**	.58	.69	.70	.71	.71	.68	.69							
skewed blobs	.64	.72	.77	.75	.79	.76	.78	.79	.79	.79	**.82**	.80	.81							
spiral	.24	.23	.43	.63	.52	.31	.51	.64	.66	**.80**	.79	.72	.73							
checkerboard	.05	-.00	.37	.75	.25	.11	.30	.74	.75	.76	.79	**.82**	.79							
20 features	.38	.36	.40	.52	**.67**	.61	.60	.64	.66	.61	.61	.62	.64							
Hastie	.02	.89	.29	.38	**.90**	.72	.69	.81	.88	.25	.25	.67	.70							
kinked border	.60	.67	.55	.56	.74	**.78**	.72	.74	.75	.66	.67	.71	.74							

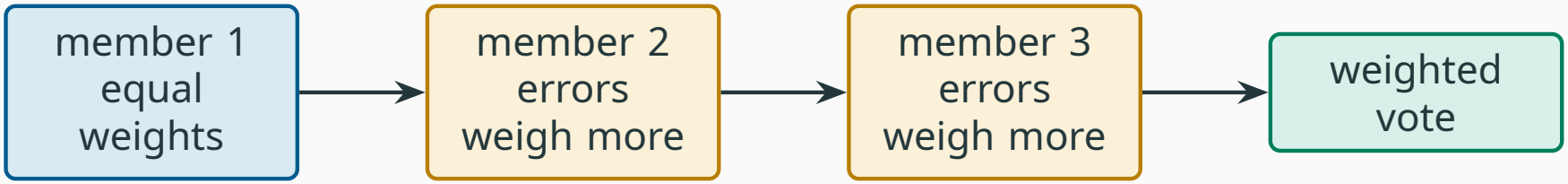
- Checkerboard: bagged trees 0.818 against 15-NN by distance 0.789
- Spiral: 0.727 at most, below 15-NN (0.802)
- Hastie: trees stay near 0.70, the SVM has 0.902

DEMO E3: `03_bagging_forests.ipynb`

Boosting

The Idea: Fix the Mistakes in Turn

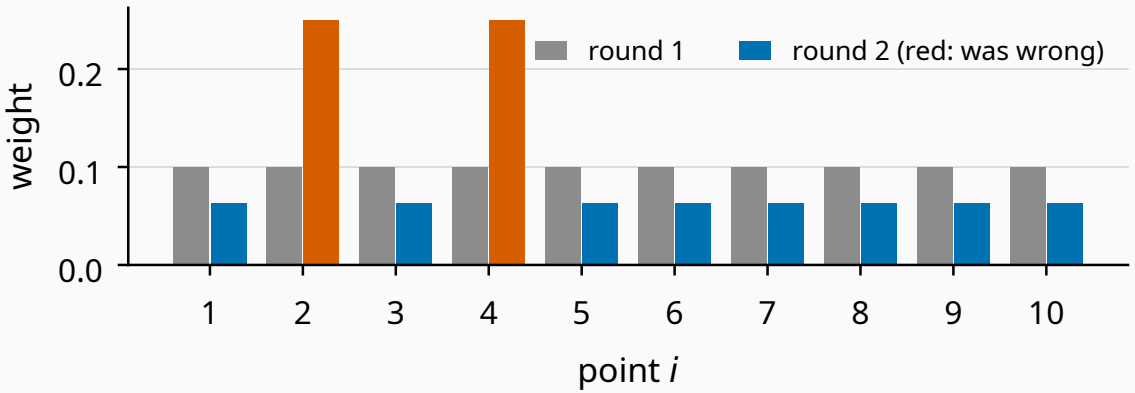
Train a weak model, find what it gets wrong, make those examples count more, train the next.



- The members are **stumps**: trees with one question (Class 06)
- Bagging cuts variance with deep trees; boosting cuts **bias** with shallow ones

AdaBoost on Ten Points: Round 1

Ten points; every weight starts at $w_i^{(1)} = 1/10$.



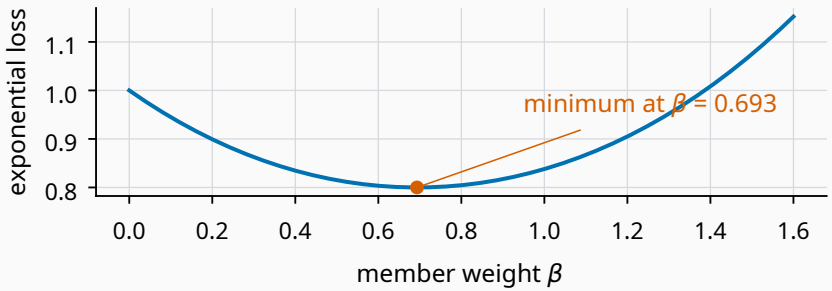
- Stump $x_1 \leq 0.053$ misses points 2 and 4: $\text{err}_1 = 0.2$
- Wrong points: $0.10 \to 0.25$; the others $0.10 \to 1/16$

The Member Weight

$$E_b = (1 - \text{err}_b)\, e^{-\beta_b} + \text{err}_b\, e^{\beta_b} \quad \text{(definition)}$$

$$e^{2\beta_b} = \frac{1 - \text{err}_b}{\text{err}_b} \quad \text{(set the derivative to zero)}$$

$$\beta_b = \frac{1}{2} \ln \frac{1 - \text{err}_b}{\text{err}_b} \quad \text{(solve for } \beta_b\text{)}$$



- $\text{err}_1 = 0.2$: $\beta_1 = \frac{1}{2} \ln 4 = 0.693$, $E_1 = 0.800$

The Weight Update

$$w_i^{(b+1)} = \frac{w_i^{(b)}\, e^{-\beta_b\, y_i\, \hat{f}_b(x_i)}}{Z_b} \quad \text{(definition)}$$

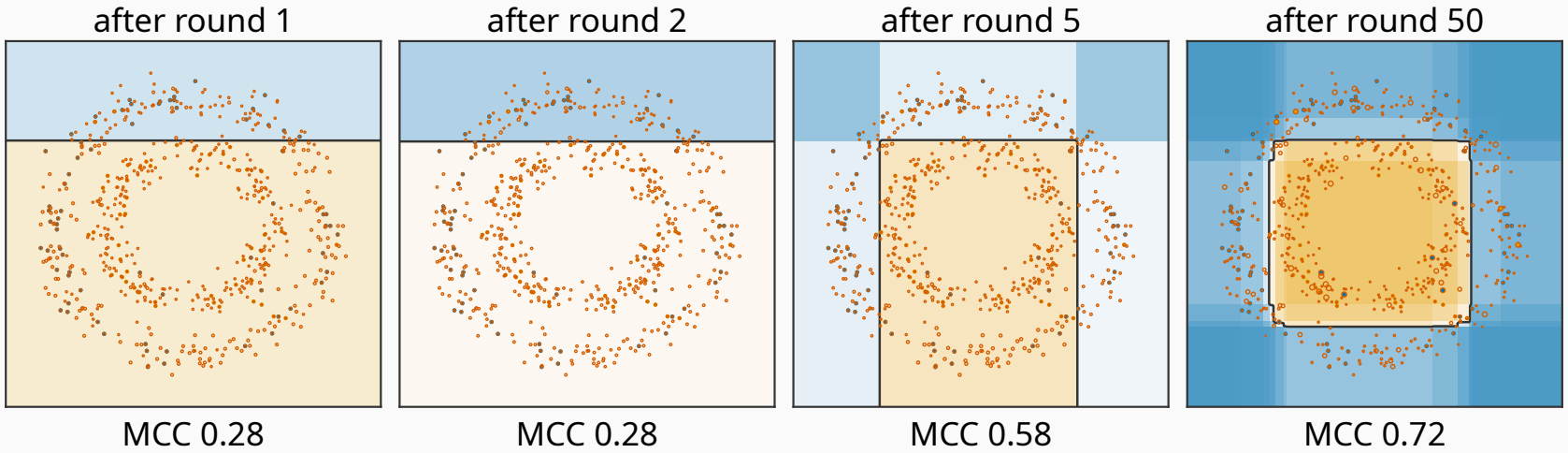
right point: weight shrinks

wrong point: weight grows

- Z_b makes the weights sum to 1; labels $y_i, \hat{f}_b(x_i) \in \{-1, +1\}$
- Round 2: $\text{err}_2 = 0.125$, $\beta_2 = 0.973$; round 3: 0.214, 0.650
- The vote sign $\sum_b \beta_b \hat{f}_b(x)$ is right on 10 of 10

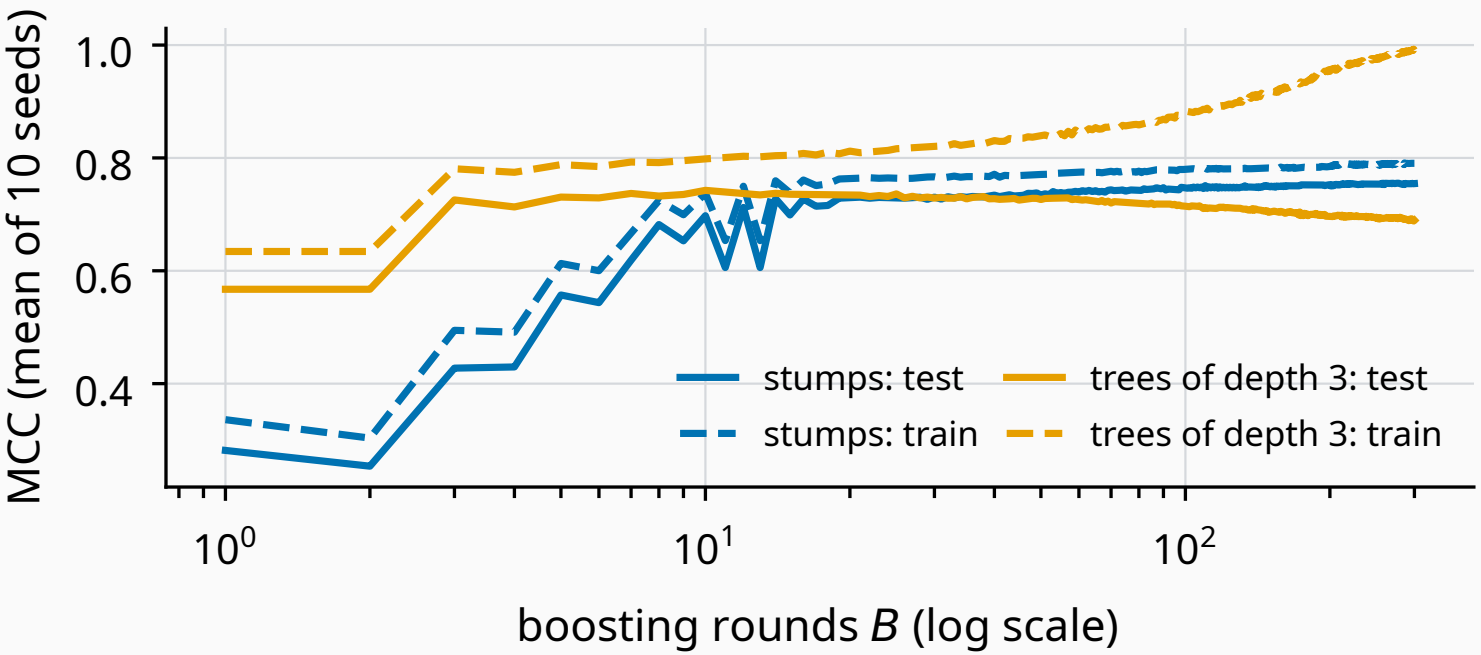
AdaBoost in Action: the Circles

Red rings: the weight of each point.



- After rounds 1, 2, 5, 50: test MCC 0.28, 0.28, 0.58, 0.72
- The largest weight reaches 4.8 times the starting $1/n$
- Hand-made and `AdaBoostClassifier` agree on every test point

More Rounds: When to Stop



- Stumps: test MCC 0.754 at 300 rounds, train 0.790
- Depth 3: best 0.743 at 10 rounds, then 0.689 with train 0.991
- Boosting **can overfit**: the rounds are a hyperparameter

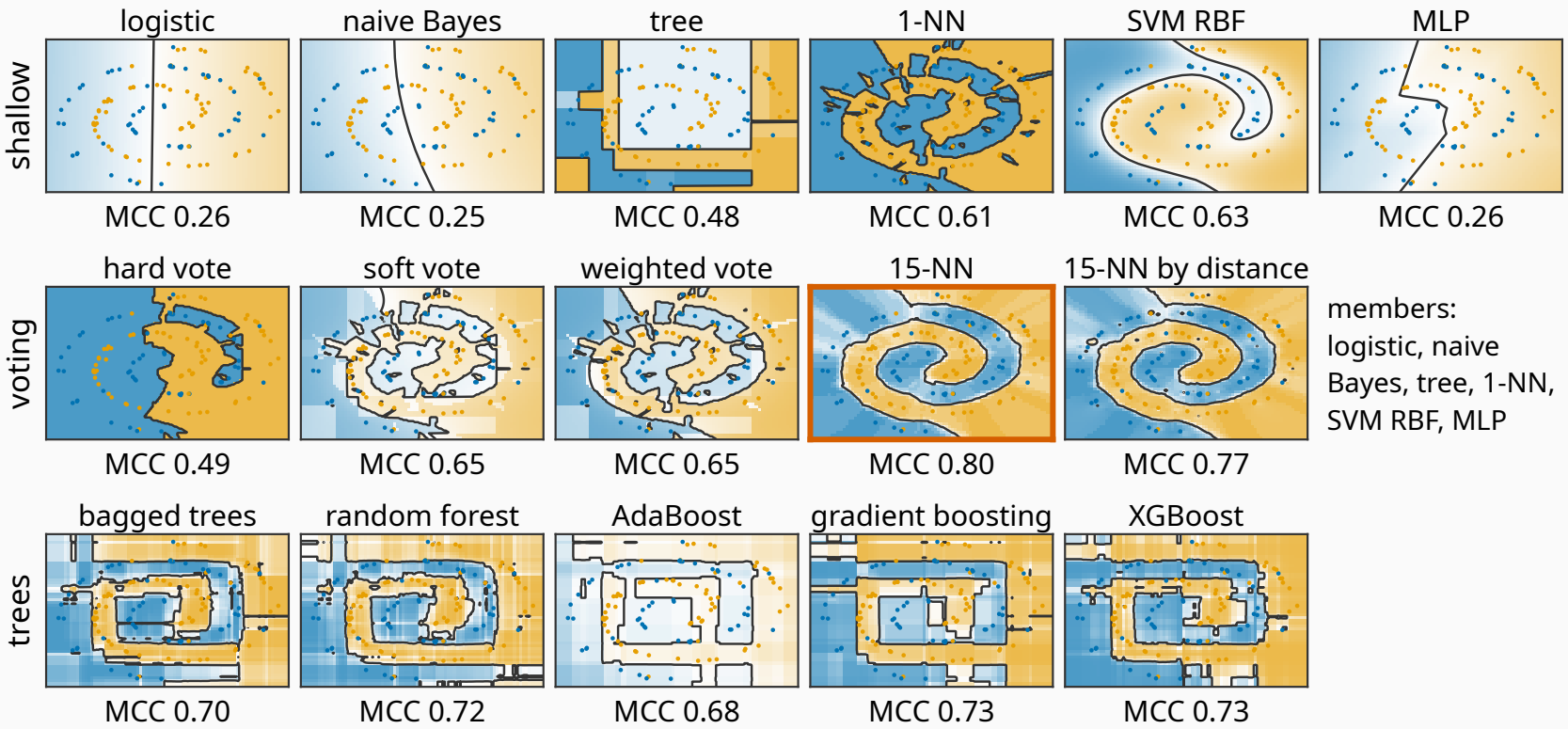
Gradient Boosting: Fit the Residuals

Think of adding a member as a **step of gradient descent in the space of functions**.

- Start from a constant F_0 (the mean)
- Squared loss: the negative gradient is the residual $r_i = y_i - F_{b-1}(x_i)$
- Fit a small tree $\hat{f}_b$ to the residuals; set $F_b = F_{b-1} + \eta \hat{f}_b$
- Eight points, $\eta = 0.5$, stumps: mean squared error (MSE) $0.461 \to 0.372 \to 0.337$

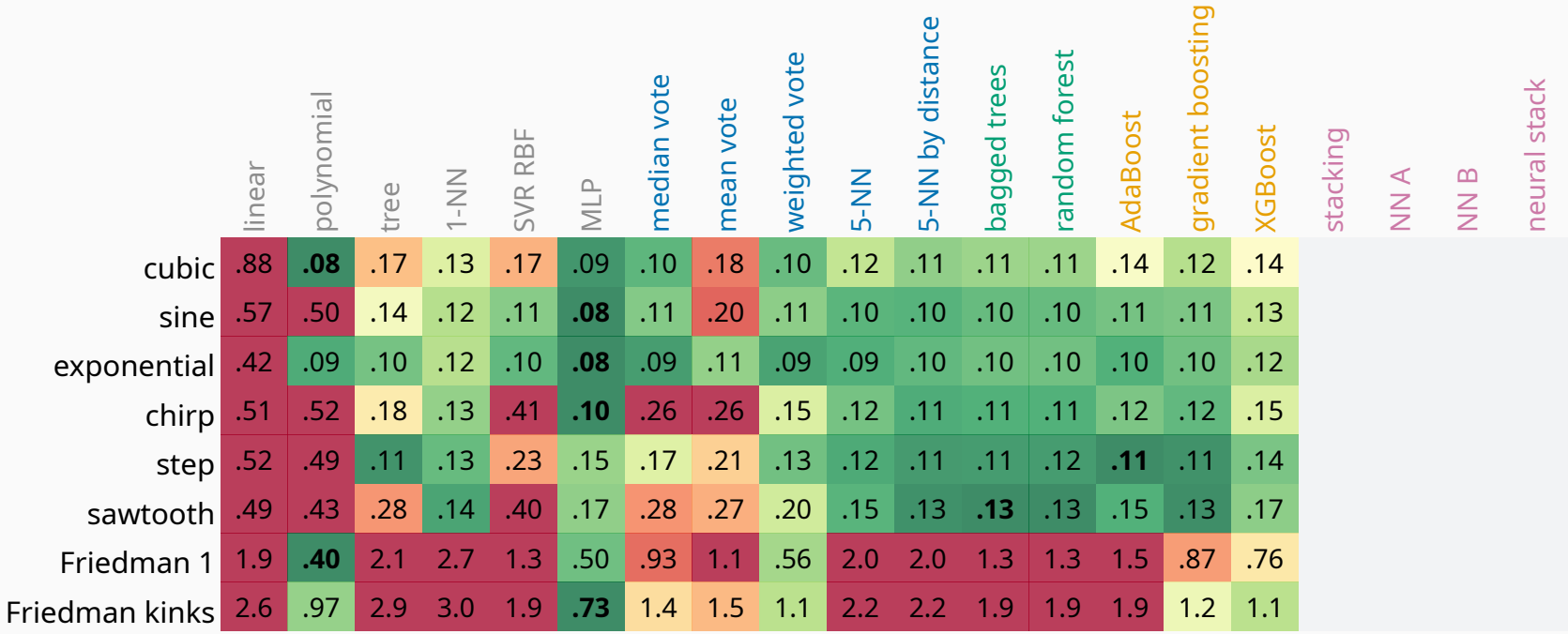
Boosting Joins the Gallery

Spiral, one seed: the row of trees now has five panels.



- Trees reach 0.68 to 0.73: staircases on a round frontier
- 15-NN stays the best (0.80)

Boosting Fills Three Columns



	linear	polynomial	tree	1-NN	SVR RBF	MLP	median vote	mean vote	weighted vote	5-NN	5-NN by distance	bagged trees	random forest	AdaBoost	gradient boosting	XGBoost	stacking	NN A	NN B	neural stack
cubic	.88	**.08**	.17	.13	.17	.09	.10	.18	.10	.12	.11	.11	.11	.14	.12	.14				
sine	.57	.50	.14	.12	.11	**.08**	.11	.20	.11	.10	.10	.10	.10	.11	.11	.13				
exponential	.42	.09	.10	.12	.10	**.08**	.09	.11	.09	.09	.10	.10	.10	.10	.10	.12				
chirp	.51	.52	.18	.13	.41	**.10**	.26	.26	.15	.12	.11	.11	.11	.12	.12	.15				
step	.52	.49	.11	.13	.23	.15	.17	.21	.13	.12	.11	.11	.12	**.11**	.11	.14				
sawtooth	.49	.43	.28	.14	.40	.17	.28	.27	.20	.15	.13	**.13**	.13	.15	.13	.17				
Friedman 1	1.9	**.40**	2.1	2.7	1.3	.50	.93	1.1	.56	2.0	2.0	1.3	1.3	1.5	.87	.76				
Friedman kinks	2.6	.97	2.9	3.0	1.9	**.73**	1.4	1.5	1.1	2.2	2.2	1.9	1.9	1.9	1.2	1.1				

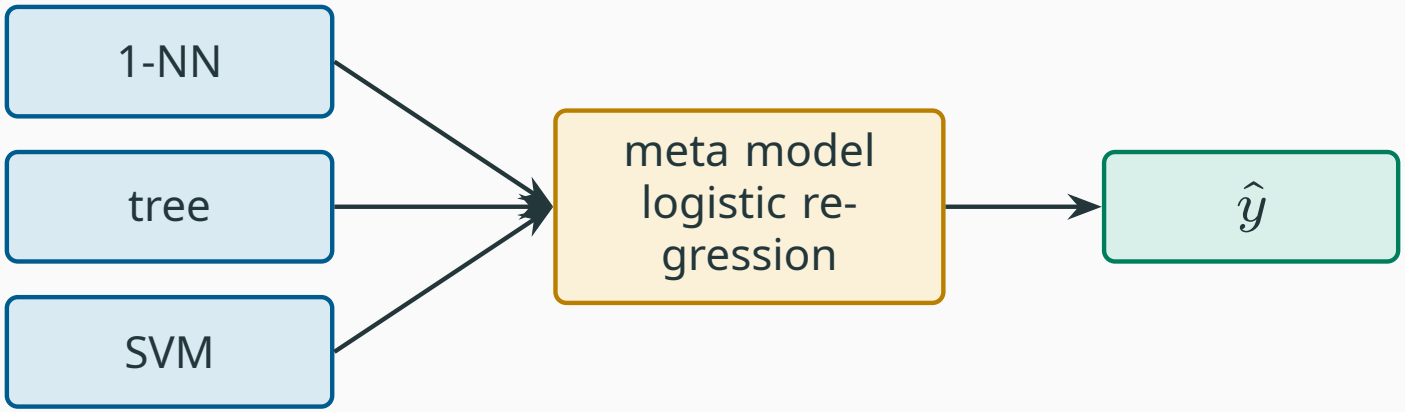
- Step: AdaBoost 0.108 wins; sawtooth: gradient boosting 0.130
- Friedman kinks: XGBoost 1.05, the MLP 0.73

DEMO E4: `04_boosting.ipynb`

Meta Models

Learn the Combination

A **meta model** takes the outputs of the members as its features.

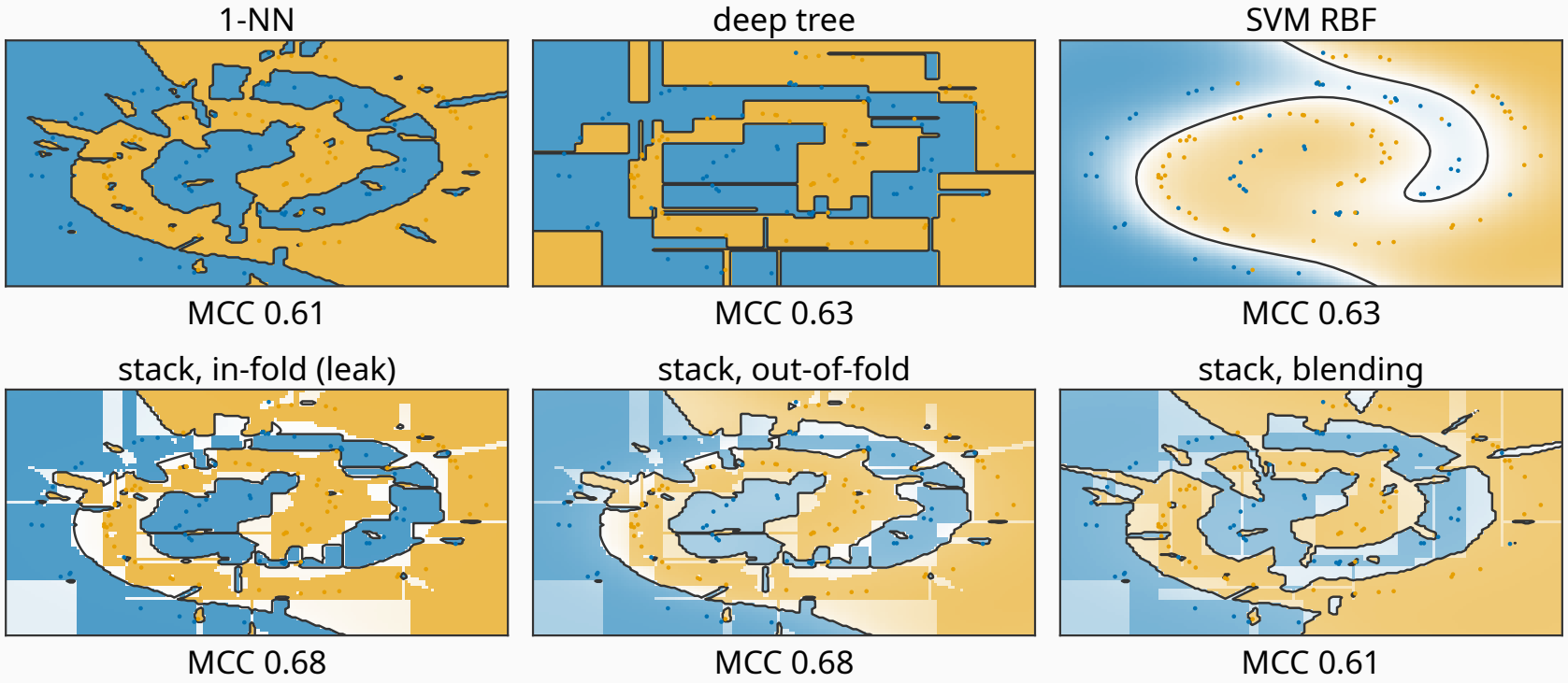


- This is **stacking**: the weights of a vote are learned, not chosen

The Leakage Trap

- A member predicts its own training points too well
- Feed those predictions to the meta model and it trusts the members too much
- **Out-of-fold predictions:** each example is predicted by a member that did not see it
- A **blend** uses one hold-out set instead

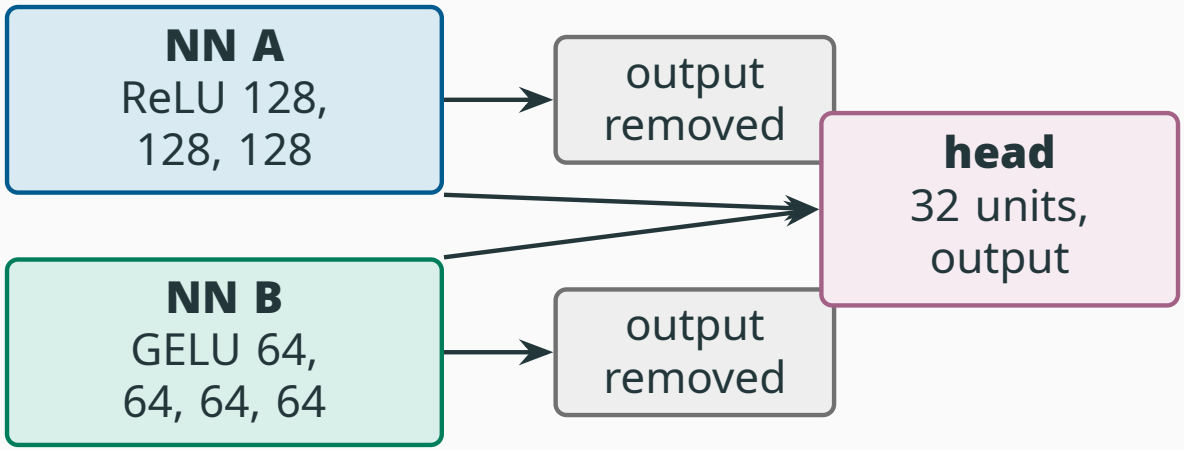
Stacking in Action: Leak or No Leak



- In-fold features: cross-validation (CV) MCC **1.000**, test 0.677
- Out-of-fold features: CV 0.650, test 0.683: an honest estimate
- Weights in-fold 4.2, 4.2, 0.65: the SVM is almost ignored

Neural Stacking: Stack What Networks Learned

Stacking reads predictions. **Neural stacking** reads the last hidden layers.



- Blue and green are **frozen**: `fit` changes only the head

The Recipe

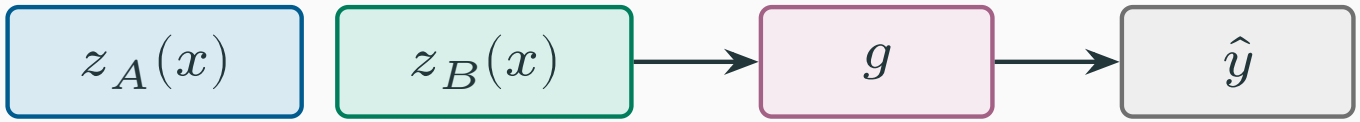
1. Build two networks of different topology
2. Train both on the task; 20 % of the data stops them early
3. Remove the output layers and freeze both
4. Concatenate the last hidden layers; add one hidden layer and an output
5. Train only this head, with the same early stopping

- Tables use tanh (256, 64) and swish (128, 128, 32)
- Regression trains on squared error and reports MAE

The Formula

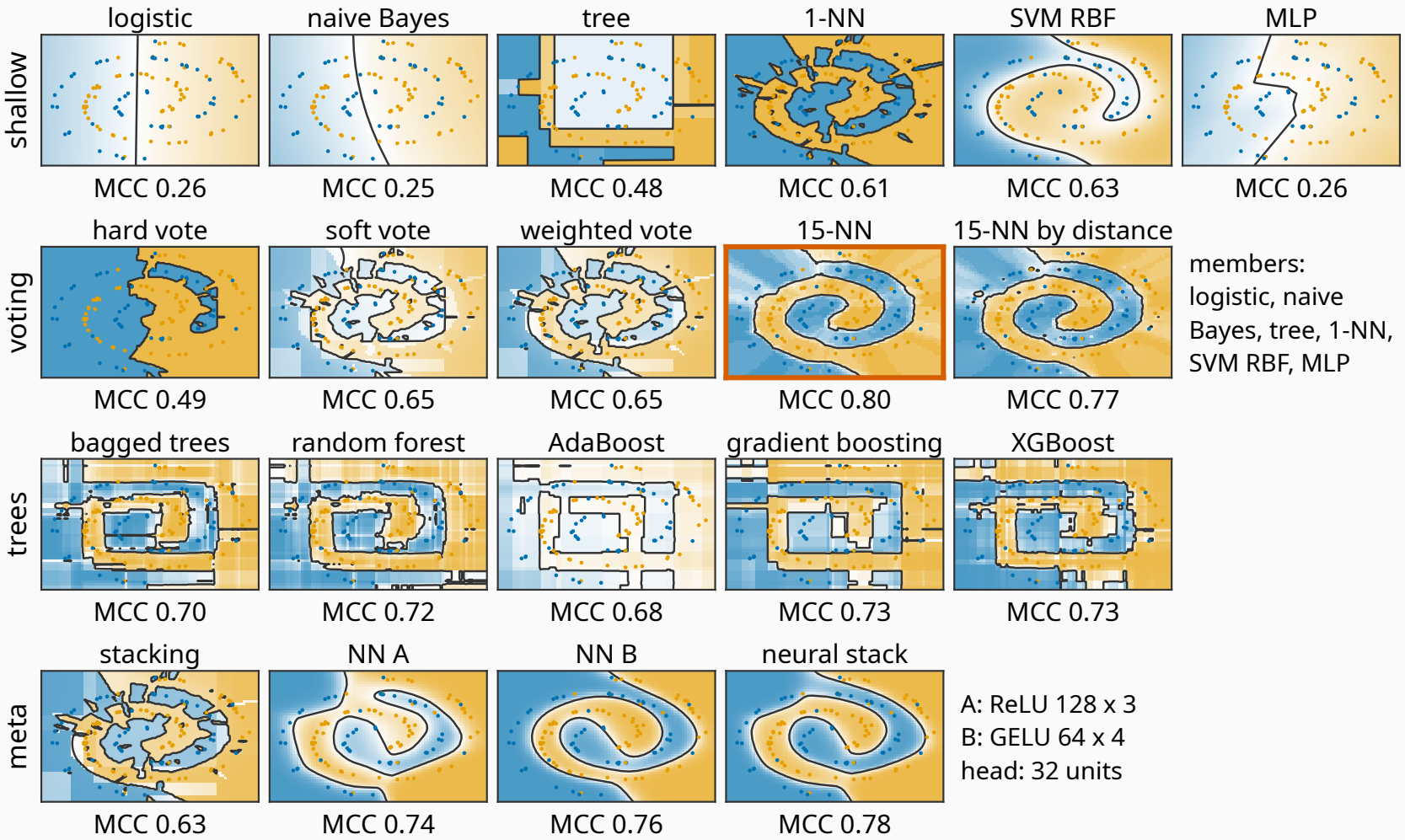
$z_A(x)$, $z_B(x)$: last hidden layers of the frozen networks. g: the head, the only trainable part.

$$\hat{y}(x) = g([\, z_A(x),\, z_B(x) \,]) \quad \text{(definition)}$$

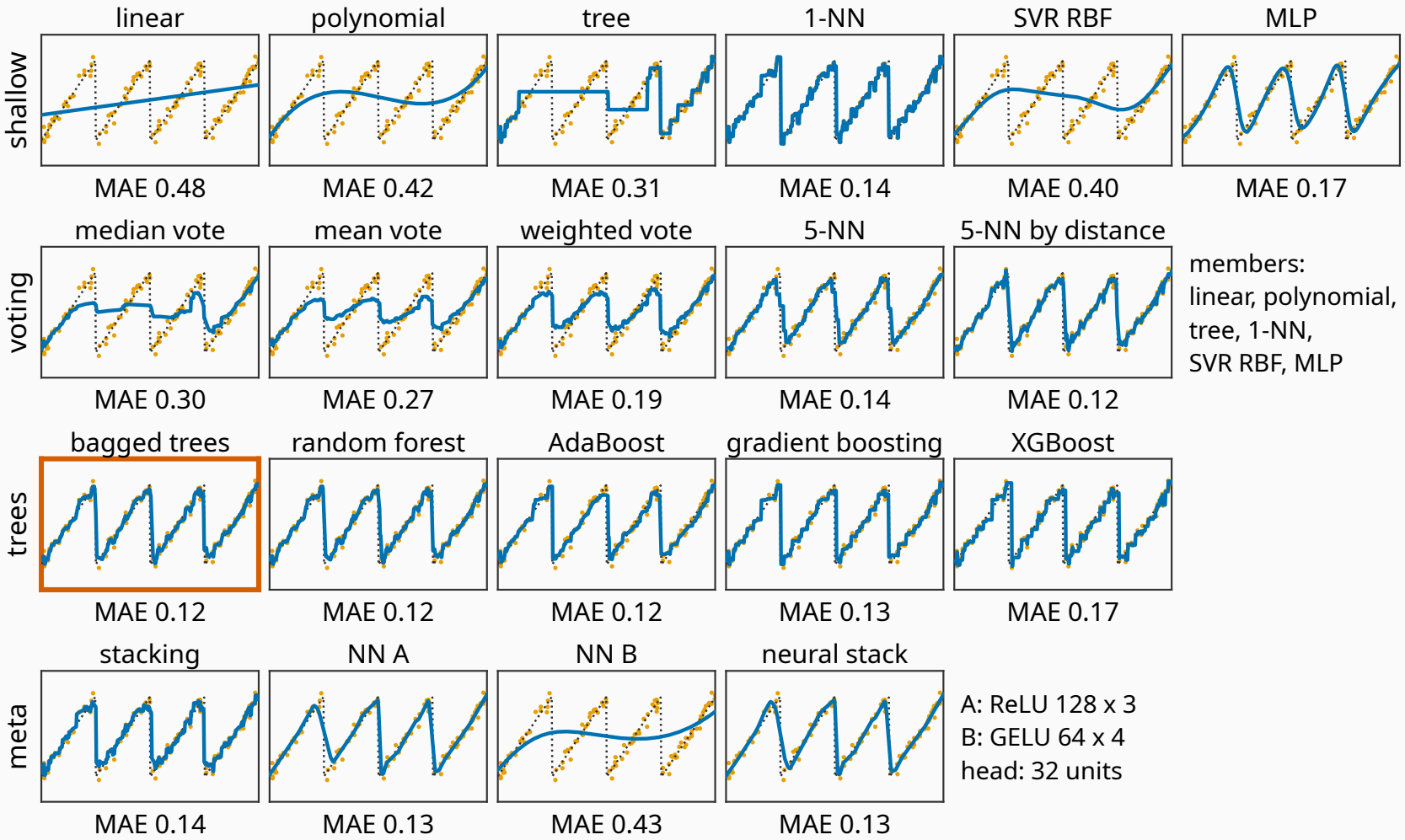


- A hidden layer is a set of features built for this task
- Two topologies give two feature sets: **diversity again**

The Complete Gallery: Spiral



The Complete Gallery: Sawtooth



The Board Is Complete

	logistic	naive Bayes	tree	1-NN	SVM RBF	MLP	hard vote	soft vote	weighted vote	15-NN	15-NN by distance	bagged trees	random forest	AdaBoost	gradient boosting	XGBoost	stacking	NN A	NN B	neural stack
moons	.60	.59	.73	.62	.76	.70	.72	.74	.74	**.78**	.77	.73	.74	.73	.74	.74	.76	.76	.76	.77
circles	.04	.74	.71	.63	**.79**	.78	.78	.77	.77	.79	.78	.71	.72	.73	.72	.72	.78	.77	.77	.78
XOR	.00	-.04	.58	.58	.71	.71	.58	.69	.70	.71	.71	.68	.69	.71	.70	.69	.71	.72	.72	**.73**
skewed blobs	.64	.72	.77	.75	.79	.76	.78	.79	.79	.79	**.82**	.80	.81	.77	.80	.79	.78	.79	.79	.79
spiral	.24	.23	.43	.63	.52	.31	.51	.64	.66	**.80**	.79	.72	.73	.73	.72	.74	.64	.77	.78	.78
checkerboard	.05	-.00	.37	.75	.25	.11	.30	.74	.75	.76	.79	**.82**	.79	.42	.57	.79	.75	.80	.80	.81
20 features	.38	.36	.40	.52	.67	.61	.60	.64	.66	.61	.61	.62	.64	.50	.55	.60	**.69**	.66	.64	.68
Hastie	.02	.89	.29	.38	.90	.72	.69	.81	.88	.25	.25	.67	.70	.73	.70	.72	**.91**	.53	.90	.91
kinked border	.60	.67	.55	.56	.74	.78	.72	.74	.75	.66	.67	.71	.74	**.82**	.77	.79	.78	.77	.80	.81

- Generated sets, defaults: voting 3, meta 3, bagging 1, boosting 1
- Neural stack: first on XOR (0.725), 0.812 on the checkerboard

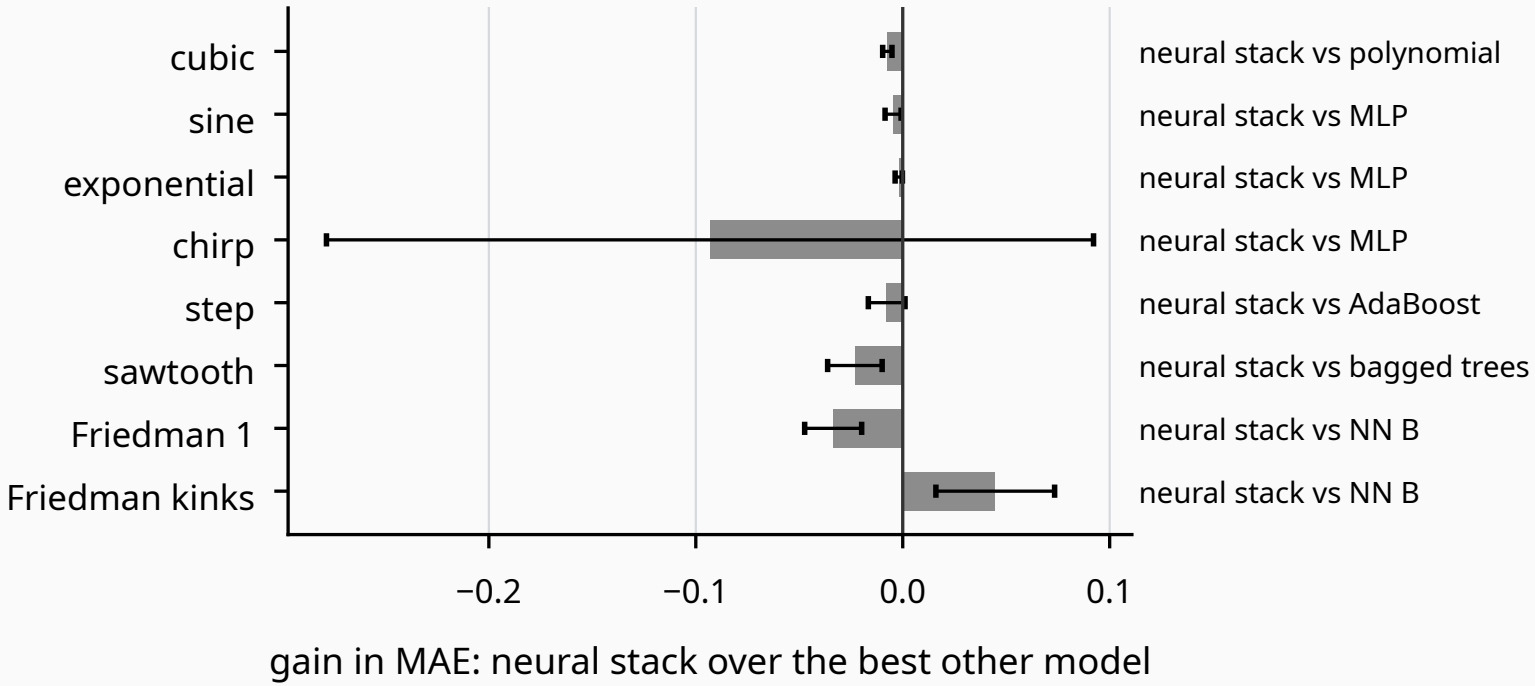
The Regression Board Is Complete

	linear	polynomial	tree	1-NN	SVR RBF	MLP	median vote	mean vote	weighted vote	5-NN	5-NN by distance	bagged trees	random forest	AdaBoost	gradient boosting	XGBoost	stacking	NN A	NN B	neural stack
cubic	.88	**.08**	.17	.13	.17	.09	.10	.18	.10	.12	.11	.11	.11	.14	.12	.14	.09	.10	.10	.09
sine	.57	.50	.14	.12	.11	**.08**	.11	.20	.11	.10	.10	.10	.10	.11	.11	.13	.09	.14	.09	.09
exponential	.42	.09	.10	.12	.10	**.08**	.09	.11	.09	.09	.10	.10	.10	.10	.10	.12	.09	.09	.09	.08
chirp	.51	.52	.18	.13	.41	**.10**	.26	.26	.15	.12	.11	.11	.11	.12	.12	.15	.10	.20	.36	.19
step	.52	.49	.11	.13	.23	.15	.17	.21	.13	.12	.11	.11	.12	**.11**	.11	.14	.11	.12	.21	.12
sawtooth	.49	.43	.28	.14	.40	.17	.28	.27	.20	.15	.13	**.13**	.13	.15	.13	.17	.14	.15	.46	.15
Friedman 1	1.9	.40	2.1	2.7	1.3	.50	.93	1.1	.56	2.0	2.0	1.3	1.3	1.5	.87	.76	.31	.40	**.30**	.34
Friedman kinks	2.6	.97	2.9	3.0	1.9	.73	1.4	1.5	1.1	2.2	2.2	1.9	1.9	1.9	1.2	1.1	.67	.59	.57	**.53**

- Smooth curves stay with the shallow MLP and the polynomial: the floor is reached
- Friedman kinks: neural stack 0.525, the best of all

Does the Neural Stack Win?

Neural stack minus the best other model, five seeds.



- Friedman kinks: $+0.045$ over NN B, won on 4 of 5 seeds
- On one input, with 120 points, simple models are better
- Friedman 1: NN B alone wins (0.304 against 0.338)

DEMO E5: `05_stacking_neural.ipynb`

Tuning

Parameters and Hyperparameters

`fit` learns the parameters. You choose the hyperparameters before `fit`.

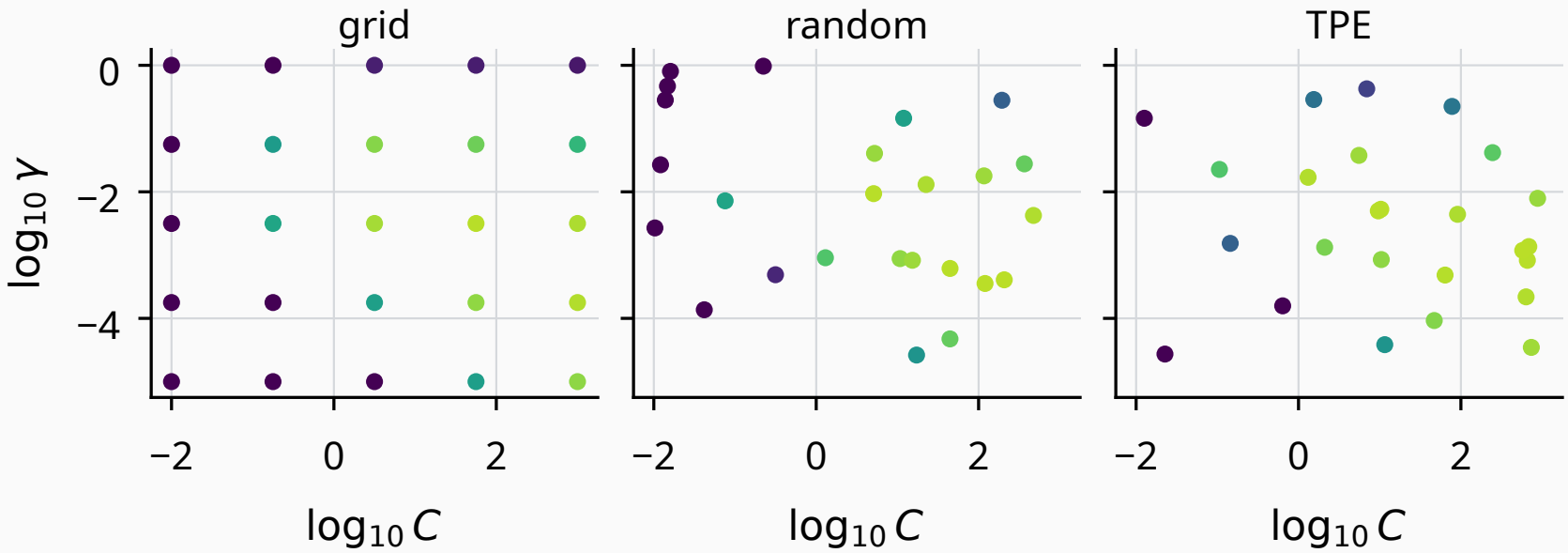
parameters
learned by `fit`:
the splits of a tree

hyperparameters
chosen first: depth, η, B, k

- Ensembles have many hyperparameters
- An untuned comparison says little: this is why the earlier boards used defaults only to find weaknesses

Three Ways to Search

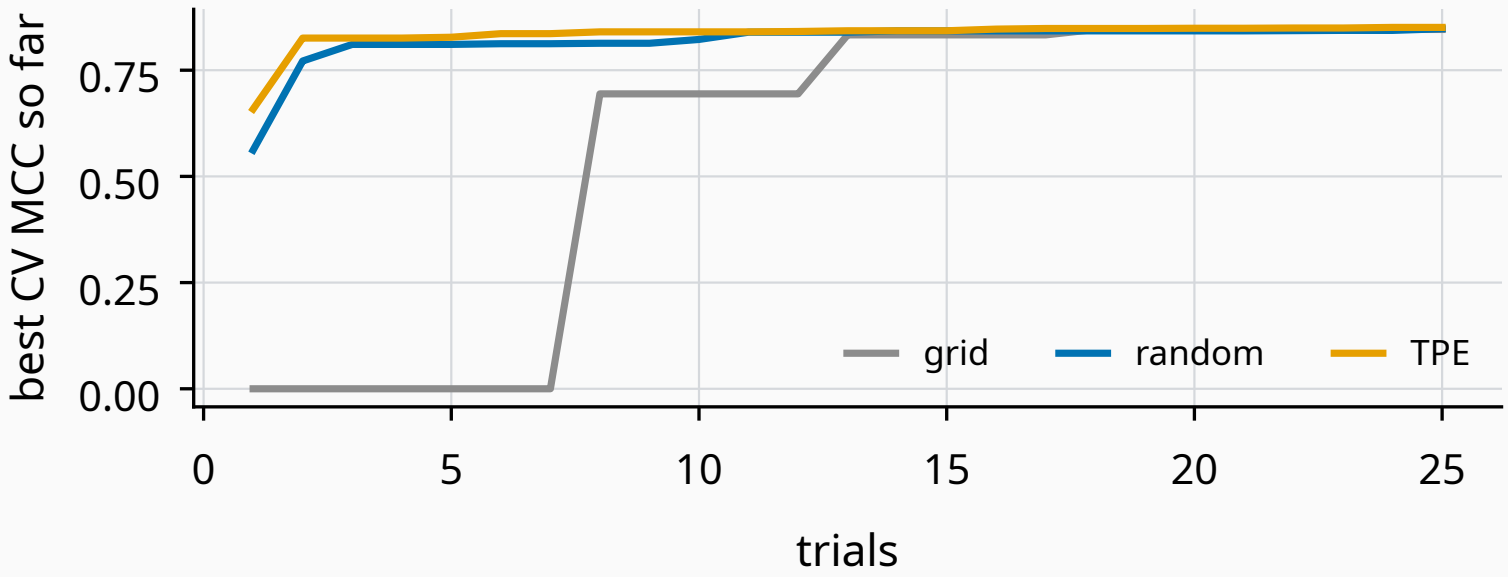
Radial basis function (RBF) SVM, 25 trials; yellow is best.



- A grid tries every combination; most points are wasted
- Random points cover each hyperparameter better (Bergstra and Bengio)
- **Tree-structured Parzen estimator** (TPE): near good trials

Learning from Past Trials

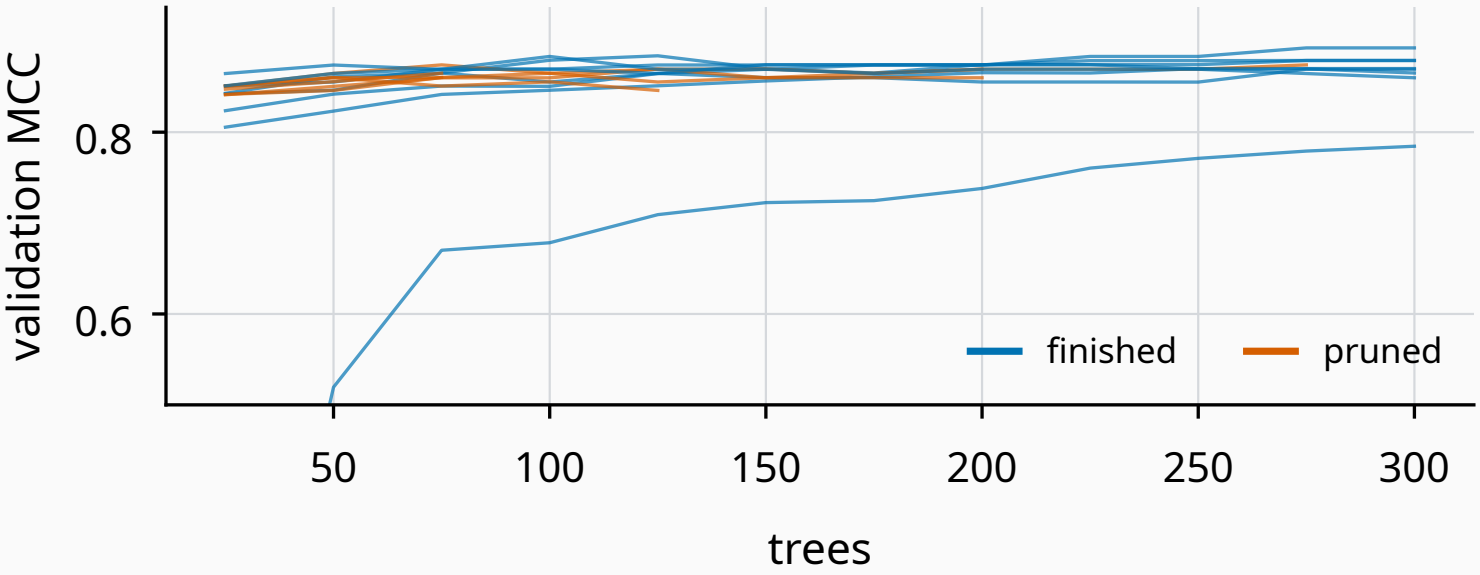
TPE splits past trials into good and bad.



- After 5 trials: grid 0.000, random 0.811, TPE 0.828
- After 25 trials: 0.846, 0.848, 0.851
- Optuna runs this search in a few lines

Pruning: Stop a Bad Trial Early

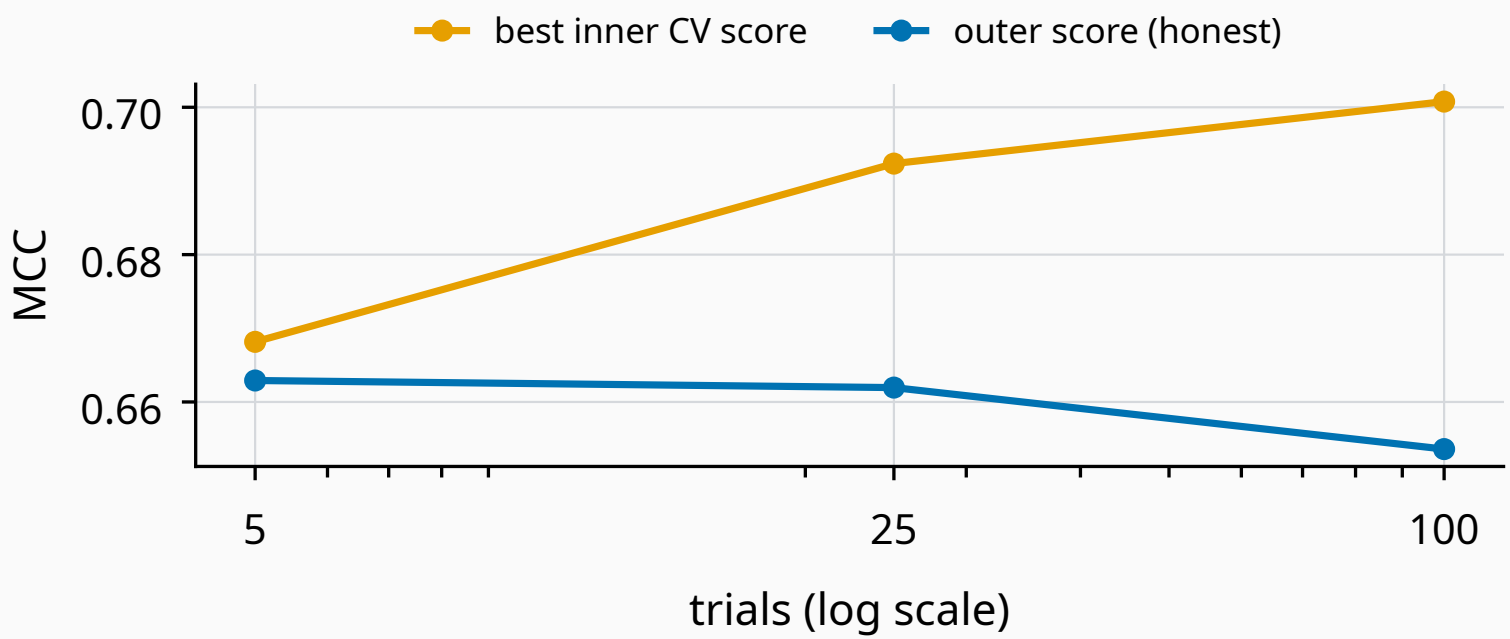
XGBoost, 30 trials. Each trial reports its score every 25 trees.



- Median pruner: stop a trial below the median of earlier trials
- Same best MCC, 0.893: 3750 trees against 9000

The Best Score Is Optimistic

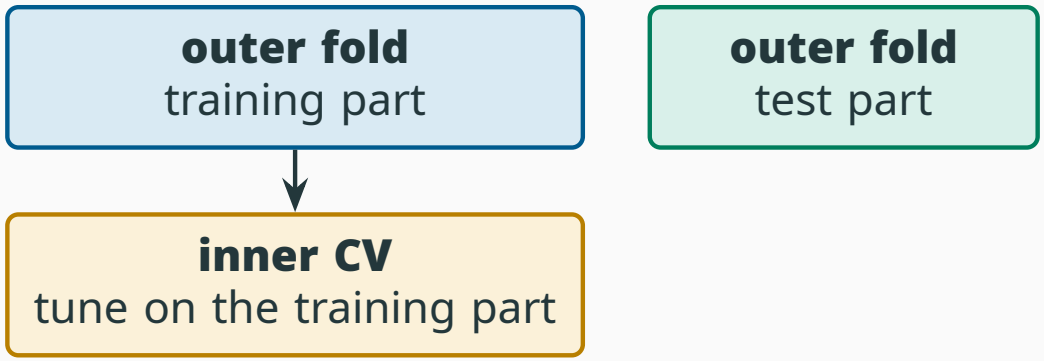
Picking the best of many trials also picks lucky noise.



- Heart disease, 303 patients: inner 0.701, honest 0.654 at 100 trials
- More trials give a larger gap: 0.005, 0.030, 0.047

Nested Cross-Validation

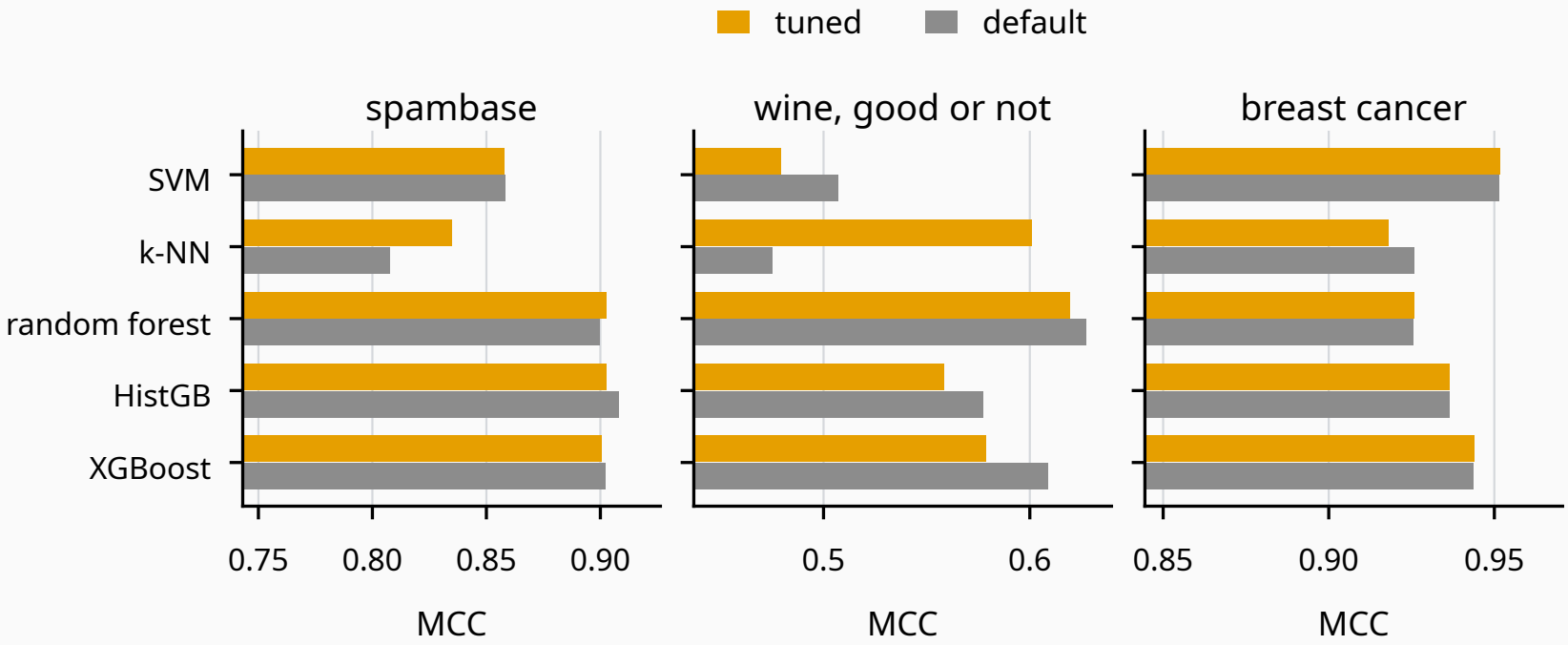
Two loops: the inner one tunes, the outer one only scores.



- The test part never touches the tuning
- The same rule as the leak in stacking

Does Tuning Pay?

Default against 25 Optuna trials, on at most 1500 rows, five folds.



- k-NN gains most: 0.475 to 0.601 on wine
- Boosting loses on wine: 0.609 to 0.579 with XGBoost
- Tune on enough data: 1500 of 6497 rows was too few

DEMO E6: `06_tuning.ipynb`

Real Tables

From Toy Data to Real Tables

Generated data showed where models break. Now: which family wins in practice?

- 11 real tables, **all rows**, 5-fold CV, the same folds for every model
- Optuna tunes on the training folds only; boosting stops early
- Stacking uses five strong members: SVM, forest, boosting, k-NN, linear

Classification on Real Tables

	linear	naive Bayes	tree	1-NN	SVM	MLP	soft vote	weighted vote	k-NN	bagged trees	random forest	AdaBoost	HistGB	XGBoost	stacking	neural stack
spambase	.84	.67	.81	.82	.87	.86	.88	.88	.83	.89	.90	.89	.90	.90	**.91**	.88
wine, good or not	.42	.30	.44	.54	.38	.46	.56	.56	.60	.63	.63	.48	.57	.58	**.63**	.51
German credit	.36	**.39**	.28	.22	.36	.35	.38	.39	.29	.37	.38	.38	.33	.35	.38	.35
breast cancer	**.96**	.87	.84	.91	.96	.85	.95	.95	.93	.91	.92	.94	.94	.94	.96	.94
heart disease	.65	.56	.54	.51	.65	.59	.67	**.68**	.65	.54	.66	.55	.65	.67	.68	.64

- Spambase: stacking 0.913, forest 0.903, one tree 0.813
- Wine: stacking 0.632, SVM 0.379
- Small, noisy tables: German credit, 0.394, leaves little to gain

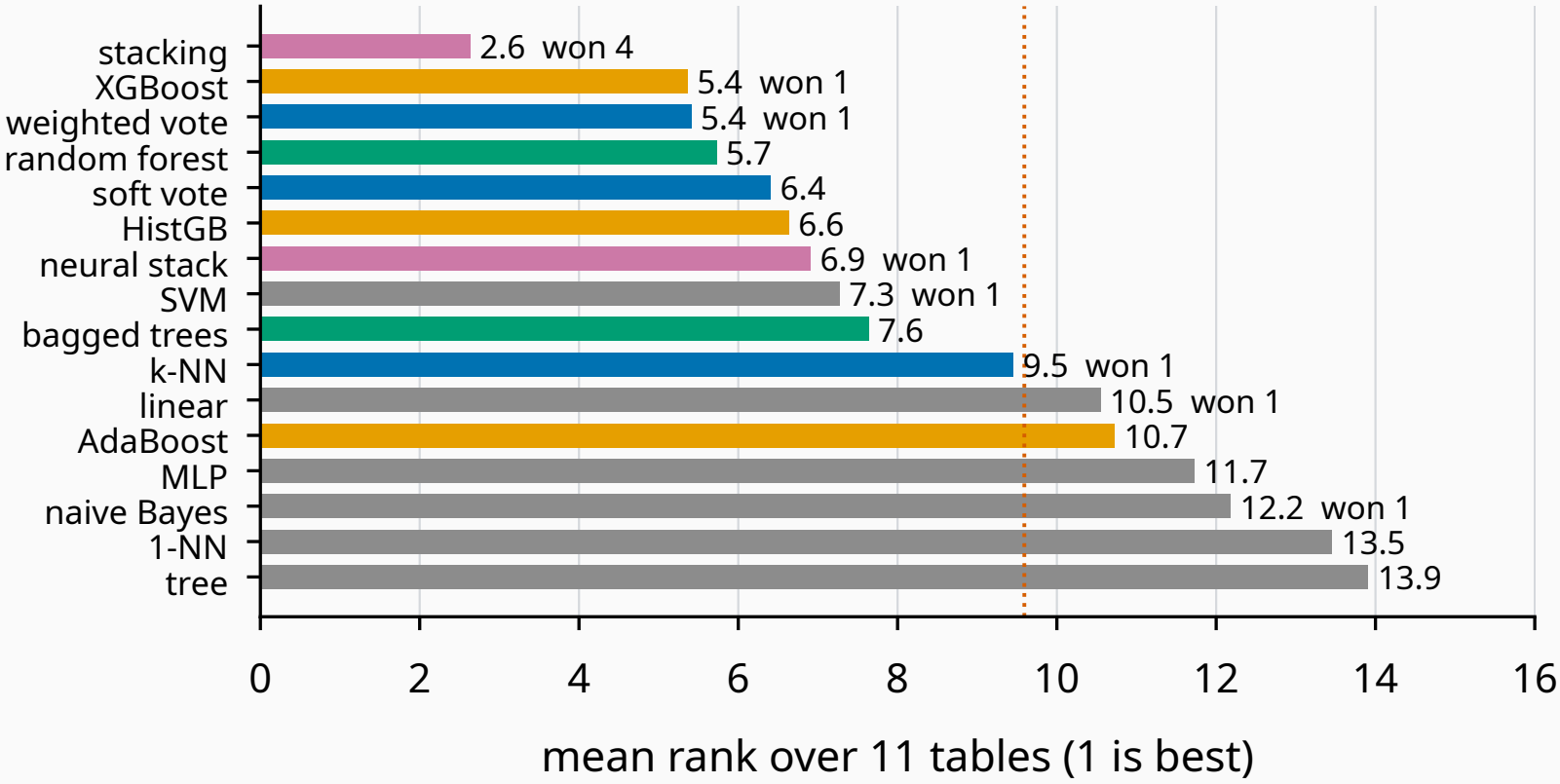
Regression on Real Tables

	linear	naive Bayes	tree	1-NN	SVM	MLP	soft vote	weighted vote	k-NN	bagged trees	random forest	AdaBoost	HistGB	XGBoost	stacking	neural stack
California housing	.49	.49	.42	.51	.37	.38	.37	.37	.41	.33	.33	.90	.30	.31	**.30**	.34
wine quality	.57	.57	.57	.45	.51	.54	.49	.49	**.40**	.43	.43	.57	.45	.41	.41	.54
abalone	1.6	1.6	1.6	2.0	**1.5**	1.6	1.5	1.5	1.5	1.6	1.6	2.2	1.5	1.5	1.5	1.5
concrete	8.3	8.3	4.4	6.3	3.8	5.1	4.3	4.4	5.7	3.4	3.4	5.5	2.9	**2.6**	2.8	3.4
diabetes	44.3	44.3	50.4	60.4	44.4	49.2	44.4	44.3	46.0	46.8	46.8	46.4	45.9	44.9	44.4	**43.0**
auto mpg	2.6	2.6	2.3	2.4	1.9	2.4	1.9	1.9	2.1	1.9	1.9	2.0	2.0	1.9	**1.8**	1.9

- California housing: stacking 0.301, HistGB 0.304, one tree 0.418
- Concrete: XGBoost 2.619, stacking 2.799, SVM 3.819
- Diabetes, 442 rows: neural stack 42.988, linear 44.270

Which Model Ranks First?

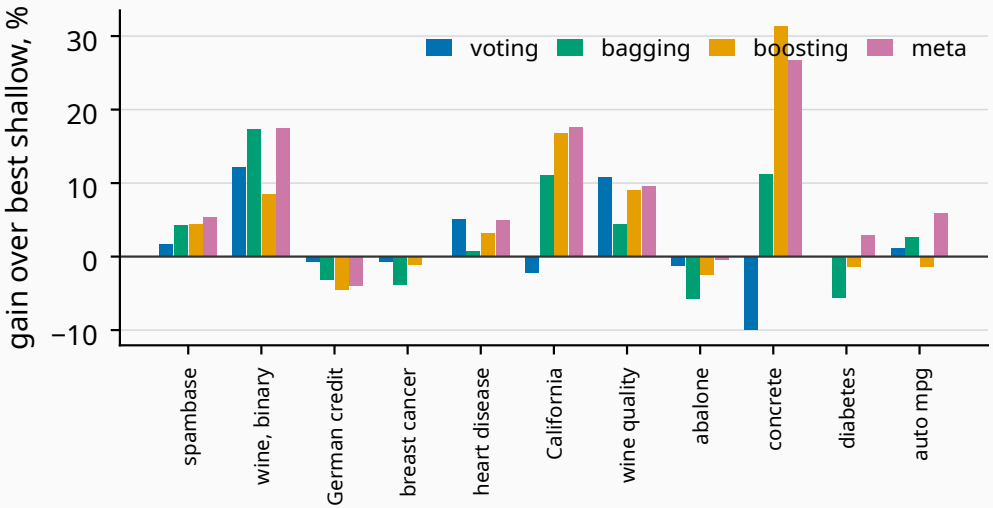
Rank the models on each table, then average; a standard comparison (Demšar).



- Friedman test $p < 10^{-9}$: the models differ
- Right of the dotted line: clearly behind stacking

How Much Does Each Family Gain?

Best model of a family against the best shallow model, on each table.



- Median gain: meta 5.4 %, boosting 3.2 %; up to 31.4 % on concrete
- Small tables gain little: more members need more data

Is the Gain Real?

McNemar: Only the Disagreements Count

Two fitted models, one test set: rows both get right say nothing (Dietterich 1998).



- Equal models: each disagreement is a coin flip
- Spambase: 11 against 13, $p = 0.839$
- MCC 0.912 and 0.915: no real gap

Which Gains Survive?

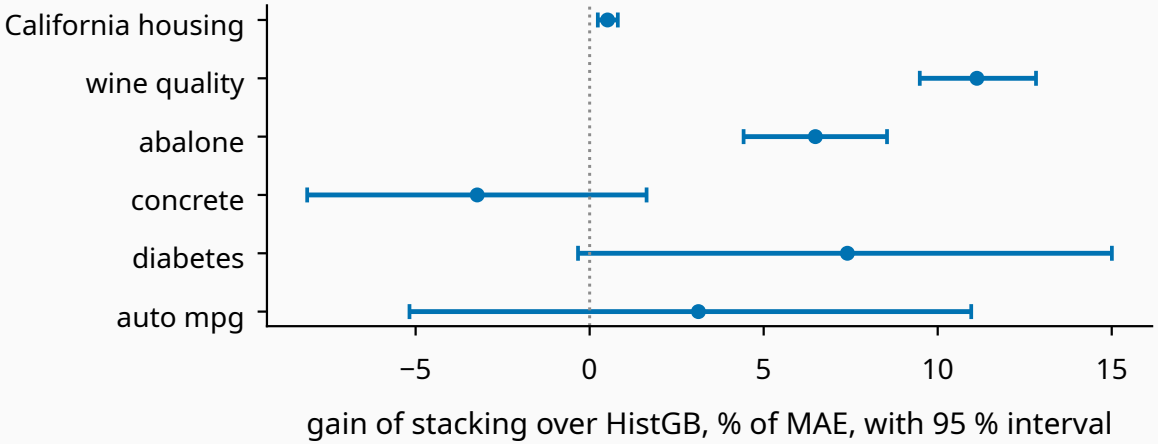
Stacking against six rivals on a 30 % hold-out, with Holm's correction per table.

	linear	tree	k-NN	SVM	random forest	HistGB
spambase	+8%	+10%	+12%	+5%	-0%	-1%
wine, good or not	+65%	+53%	+4%	+23%	+1%	+8%
German credit	-7%	+213%	+110%	-12%	-2%	+17%
breast cancer	+1%	+10%	+3%	+4%	+4%	-3%
heart disease	-10%	+43%	+0%	+23%	-0%	+4%
California housing	+37%	+33%	+24%	+15%	+9%	+1%
wine quality	+25%	+26%	-1%	+17%	+4%	+11%
abalone	+7%	+10%	+3%	+0%	+5%	+6%
concrete	+60%	+45%	+52%	+29%	+13%	-3%
diabetes	+1%	+22%	+1%	+16%	+7%	+7%
auto mpg	+27%	+25%	+12%	+5%	+4%	+3%

- Green: permutation test on the metric: swap predictions on random rows
- Better in 35 of 66, never worse; forest and HistGB fall on 4 of 11 tables
- McNemar agrees on 29 of 30: it counts errors, not MCC

Regression: No Right or Wrong

No 2×2 table exists: McNemar's twin, the sign test, asks which model came **closer**.



- Sign test: closer more often; Wilcoxon: smaller by rank; permutation: lower MAE
- Wine quality: k-NN is closer on 1133 of 1950 rows, yet MAE ties ($p = 0.138$)
- California: $p = 0.001$ for a 0.51 % gain: real, but tiny

McNemar or Friedman?

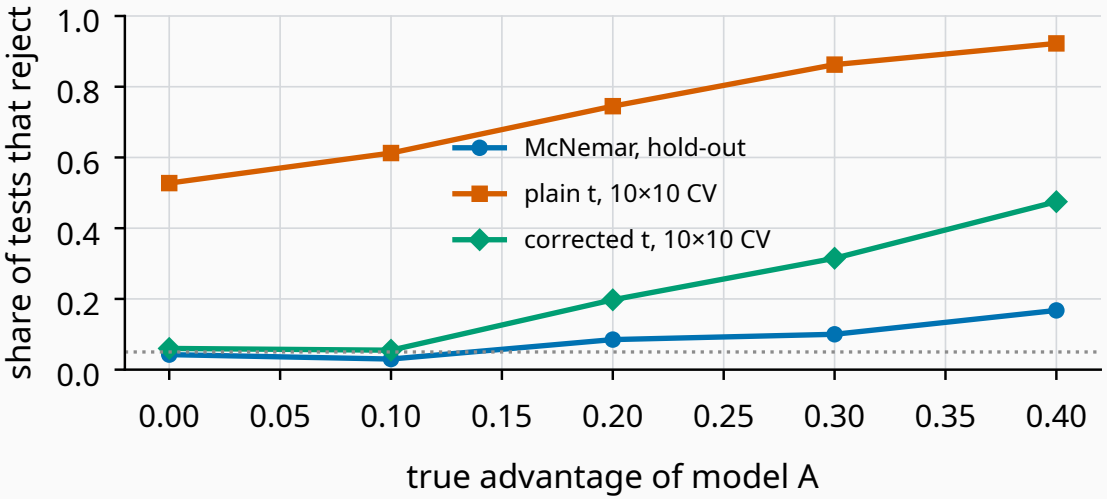
Neither is better: one family of tests, at two levels (Demšar 2006).

Samples	Two models	Many models
rows, right or wrong	McNemar	Cochran's Q
rows, loss of each row	sign test, Wilcoxon	Friedman on rows
tables, one score each	sign test, Wilcoxon	Friedman on tables

- Spambase: Cochran's Q and Friedman on rows both give 124.9
- McNemar: better **here**? Friedman: better **across tables**?

Do Not Test on the Folds

Folds share training rows; at 0.00 neither model is better (Nadeau and Bengio 2003).



- No true gap: plain t-test rejects 0.528, corrected 0.06, McNemar 0.042
- At 0.40 the corrected test finds 0.475, McNemar 0.168

Across Tables: After Friedman

A mean-rank pairwise test depends on the other models in the pool (Benavoli et al. 2016).

- Wilcoxon with Holm: 11 of 15 models behind stacking; Nemenyi: 6
- Stacking was picked **after** E7: this correction does not pay for that
- All 120 pairs with 11 tables: smallest possible Holm p is 0.117
- Bayesian signed-rank: P(better by over 1 %) at least 0.967 (Benavoli et al. 2017)

DEMO E7: `07_real_tables.ipynb`

Then `08_significance.ipynb`: the tests of this section.

Summary

Summary: the Map

	voting	bagging	boosting	meta
idea	combine outputs	average resamples	fix errors in turn	learn the mix
cuts	variance	variance	bias	both
costs	keeps bad members	memory	tuning, noise	time, leakage

Key Takeaways: Ensembles

- A crowd helps only if members are **better than chance and different**
- On real tables stacking ranks first, then boosting, forests and votes
- Averaging leaves a floor ρv: lower ρ with diverse members
- k-NN with $k > 1$ is a vote; bagging needs unstable learners
- Boosting needs a stopping rule; stacking needs out-of-fold predictions
- Tune with nested CV: the best validation score is optimistic
- Test gains: McNemar on one test set, permutation for MCC and MAE
- Near the ceiling no ensemble wins

Lab 07: Build It from Scratch

Part	Task	twin
A	Boundary gallery and a vote of three models	`VotingClassifier`
B	Bagging by hand: bootstrap, vote, OOB	`BaggingClassifier`, `RandomForestClassifier`
C	AdaBoost with stumps	`AdaBoostClassifier`
D	Gradient boosting, shrinkage, early stopping	`GradientBoostingRegressor`
E	Stacking with out-of-fold predictions	`StackingClassifier`
F	Challenge: pseudo-residuals with `jax.grad`	`HistGradientBoostingClassifier`

- **In class:** Parts A to D (60 minutes); **at home:** Parts E and F
- Guide `practice/07_ensembles.pdf`; notebook `lab07_ensembles.ipynb`

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